LANGSAMP: Language-Script Aware Multilingual Pretraining

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Abstract

Recent multilingual pretrained language models (mPLMs) often avoid using language embeddings – learnable vectors assigned to individual languages. However, this places a significant burden on token representations to encode all language-specific information, which may hinder language neutrality. To address this limitation, we propose **Lang**uage-**S**cript **A**ware **M**ultilingual **P**retraining (**LANGSAMP**), a method that incorporates both **language** and **script** embeddings to enhance representation learning. Specifically, we integrate these embeddings into the output of the Transformer blocks before passing the final representations to the language modeling head for prediction. We apply LANGSAMP to the continual pretraining of XLM-R (Conneau et al., 2020) on a highly multilingual corpus covering more than 500 languages. The resulting model consistently outperforms the baseline in zero-shot crosslingual transfer across diverse downstream tasks. Extensive analysis reveals that language and script embeddings capture language- and script-specific nuances, which benefits more language-neutral representations, proven by improved pairwise cosine similarity. In our case study, we also show that language and script embeddings can be used to select better source languages for crosslingual transfer. We make our code and models publicly available at https://github.com/cisnlp/LangSAMP.

1 Introduction

Encoder-only mPLMs are often regarded as universal text encoders (Cer et al., 2018; Huang et al., 2019; Yang et al., 2020), where the sentence-level or token-level representations are applied to various downstream tasks across different languages (Wei et al., 2021). One of the most attractive aspects of these representations is their utility in crosslingual transfer (Zoph et al., 2016; Wu and Dredze, 2019; Artetxe et al., 2020a). That is, representations from a single source language can be used to fine-tune a multilingual task-specific model (e.g., an mPLM + a task-specific classifier). The fine-tuned model can be applied directly to other languages, without further training. Such a pipeline is particularly useful for low-resource languages, where training data is often scarce (Artetxe et al., 2020b).

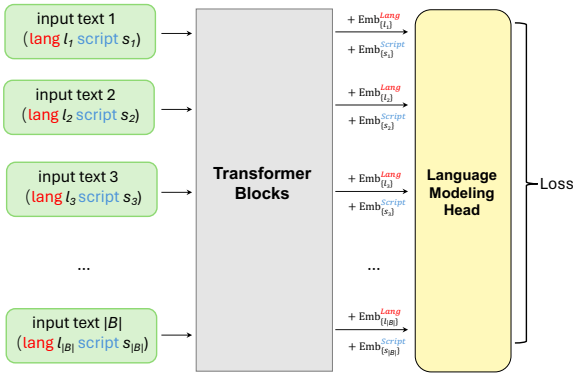


Figure 1: An illustration of LANGSAMP for a single batch. Each text may come from different languages and different scripts. Language and script embeddings are added to the transformer output before feeding into the language modeling head. This setup improves the language neutrality of the representations as the auxiliary embeddings share the burden by encoding some language- and script-specific information useful for decoding specific tokens in masked language modeling.

The effectiveness of this pipeline depends on the transferability of crosslingual representations. However, previous studies have shown that the representations from recent mPLMs encode a lot of language- and script-specific information (Datta et al., 2020; Chang et al., 2022; Wen-Yi and Mimno, 2023). This is generally not advantageous, as language neutrality, i.e., representations from different languages share a unified subspace, is important for effective crosslingual transfer (Libovický et al., 2020; Chang et al., 2022; Hua et al., 2024). While

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Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers), pages 1743–1770
July 27 - August 1, 2025 ©2025 Association for Computational Linguistics

some approaches attempt to post-align these representations (Cao et al., 2020; Pan et al., 2021; Liu et al., 2024b; Xhelili et al., 2024), limited efforts have focused on enhancing language neutrality from the architectural perspective of mPLMs during pretraining.

Early mPLMs, such as XLM (Conneau and Lample, 2019), leverage language embeddings – learnable vectors assigned to different languages. These embeddings are added to the token embeddings before being fed into the transformer (Vaswani et al., 2017) blocks, aiming to alleviate the burden of encoding language-specific information within the token embeddings. Language embeddings can also guide generation toward the correct target language in machine translation (Conneau and Lample, 2019; Song et al., 2019; Liu et al., 2022). However, more recent mPLMs, such as XLM-R (Conneau et al., 2020) and mBERT (Devlin et al., 2019), have discarded these embeddings. The two primary reasons are that (1) mPLMs are expected to have a single, unified parameter set for all languages, and (2) they need to function seamlessly as universal text encoders without requiring language IDs as input. However, the removal inevitably reduces the language neutrality of token embeddings and representations (contextual token embeddings), which may negatively impact crosslingual transfer.

To address this limitation, this work proposes **Lang**uage-**S**cript **A**ware **M**ultilingual **P**retraining (**LangSAMP**), a method that incorporates both **language** and **script** embeddings to facilitate better representation learning. Instead of adding these embeddings to the token embeddings before feeding them into the transformer blocks, we add them to the output of the transformer blocks (final contextual token embeddings) **before feeding them into the language modeling head**, as shown in Figure 1. In the pretraining phase, language and script IDs are required to obtain language and script embeddings, offloading the burden and helping decode specific tokens in masked language modeling. After pretraining, the backbone (token embeddings and transformer blocks) can function seamlessly as a universal text encoder, which can be fine-tuned together with a task-specific classifier for downstream tasks, without any language or script IDs as input, which are the same as most recent mPLMs.

To validate our approach, we continually pretrain XLM-R (Conneau et al., 2020) using LangSAMP on Glot500-c (ImaniGooghari et al., 2023), a multilingual dataset containing over 500 languages. We evaluate the resulting model across a diverse set of downstream tasks, including sentence retrieval, text classification, and sequence labeling, consistently achieving superior performance compared to the baseline. We show that better language neutrality is achieved – LangSAMP improves the pairwise cosine similarity across languages. Additionally, we observe that language and script embeddings encapsulate typological features, making their similarities a useful resource for selecting optimal source languages in crosslingual transfer.

Our main contributions are as follows: (i) We propose LangSAMP, an effective multilingual pretraining method to improve the language neutrality of representations. (ii) We conduct extensive experiments across a spectrum of downstream tasks, demonstrating that our method consistently improves crosslingual transfer performance. (iii) Our case study shows that language embeddings, as a byproduct, can effectively assist in selecting the optimal source language for crosslingual transfer.

2 Related Work

2.1 Multilingual Pretrained Language Models

Multilingual pretrained language models (mPLMs) are models that are trained on many languages, with one or multiple self-supervised objectives, such as masked language modeling (MLM) (Devlin et al., 2019) or causal language modeling (Radford et al., 2019). These models can be generally classified as encoder-only (Devlin et al., 2019; Conneau et al., 2020; Liang et al., 2023), encoder-decoder (Liu et al., 2020; Fan et al., 2021; Xue et al., 2021), and decoder-only models (Lin et al., 2022; Shliazhko et al., 2022; Scao et al., 2022). Decoder-only models that have considerably many parameters and are pretrained on a lot of data are also referred to as large language models (LLMs) (Achiam et al., 2023; Touvron et al., 2023; Üstün et al., 2024), which are good at natural language generation tasks, typically for high- and medium-resource languages. In parallel, some recent encoder-only models attempt to scale *horizontally*, i.e., cover more languages, especially low-resource ones (Ogueji et al., 2021; Alabi et al., 2022; ImaniGooghari et al., 2023; Liu et al., 2024a). These highly multilingual encoder-only models are particularly good at understanding tasks in a zero-shot crosslingual fashion.

2.2 Language Embeddings

Language embeddings are vectors that explicitly or implicitly capture the linguistic characteristics of languages. Early works construct such embeddings using prior knowledge of the languages, resulting in vectors where each dimension encodes a specific linguistic feature (Östling, 2015; Ammar et al., 2016; Littell et al., 2017). However, such features have to be manually defined and may be unavailable for less-studied languages (Yu et al., 2021). Therefore, researchers also explore learning language embeddings directly from parallel corpora (Malaviya et al., 2017; Östling and Tiedemann, 2017; Bjerva and Augenstein, 2018; Tan et al., 2019; Liu et al., 2023; Chen et al., 2023) or monolingual corpora (Conneau and Lample, 2019; Yu et al., 2021). This is usually done by assigning an ID to each language, initializing a fixed-length learnable vector, and integrating the vector into the input from that language. The embeddings can capture linguistic features and help crosslingual tasks, e.g., guiding language-specific generation in machine translation in XLM (Conneau and Lample, 2019). This line of approaches requires language IDs as input for both pretraining and downstream fine-tuning. In contrast, language embeddings are only leveraged in our pretraining. The backbone can be used as a universal text encoder without language IDs for fine-tuning on downstream tasks.

3 Methodology

We present **LANGSAMP**, an approach that incorporates both **language** and **script** embeddings to facilitate learning more language-neutral representations in multilingual pretraining. LANGSAMP preserves the same architecture as the most recent multilingual encoder-only models, except for requiring auxiliary language and script IDs/embeddings in pretraining. In the fine-tuning stage, these auxiliary IDs and embeddings are not required. We introduce the key components in the following.

3.1 Language and Script Embeddings

Language and script embeddings are introduced to share the token representations' burden of encoding language- and script-specific information. Let $\boldsymbol{E}^{Lang} \in \mathbb{R}^{L \times D}$ and $\boldsymbol{E}^{Script} \in \mathbb{R}^{S \times D}$ be the language and script embeddings, respectively, where L is the number of languages, S is the number of scripts, and D is the embedding dimension of the model. We use $\boldsymbol{E}_l^{Lang}$ (resp. $\boldsymbol{E}_s^{Script}$) to denote the embedding of a specific language l (resp. script s). Similar to token embeddings (which represent relations between tokens in vector space), the language/script embeddings are also expected to capture structural and typological similarities of languages (§5.2) and be useful for selecting good source language for crosslingual transfer (§5.4).

3.2 Language-Script Aware Modeling

In the standard MLM pretraining, Transformer blocks generate the final representation at a masked position. Subsequently, this representation is fed to the language modeling head to reconstruct the original token. Since the original token is used by a specific language and written in a specific script, language- or script-specific information is particularly necessary to decode this token. From this perspective, the Transformer output used for decoding is not language-neutral by nature. Our intuition is that we can ease the decoding by giving hints (e.g., the token should be generated in a specific language or script) to the language modeling head. In this way, the output of the Transformer blocks does not need to encode much language- and script-specific information, and can thus be more language-neutral. Inspired by this, we add language and script embeddings to the output of Transformer blocks and feed the resulting representations to the language modeling head for decoding, as shown in Figure 2.

Formally, let a training instance (an input sentence) be $X = [x_1, x_2, \cdots, x_n]$ that comes from language l and is written in script s. We feed X into Transformer blocks and obtain the final contextualized embeddings from the last layer: $\boldsymbol{H} = [\boldsymbol{h}_1, \boldsymbol{h}_2, \cdots, \boldsymbol{h}_n]$. We then add the language and script embedding to these outputs to form the final representations: $\boldsymbol{o}_i = \boldsymbol{h}_i + \boldsymbol{E}_l^{Lang} + \boldsymbol{E}_s^{Script}$. The final representations at the masked positions are used to decode the original tokens in MLM:

$$\mathcal{L}_{MLM} = -\sum_{i \in \mathcal{M}} \log P_{MLM}(x_i | \boldsymbol{o}_i)$$

where $\mathcal{M}$ is the set of masked positions in X and $P_{MLM}(x_i|\boldsymbol{o}_i)$ is the probability of decoding the original token x_i given the final representation $\boldsymbol{o}_i$, which is computed by the language modeling head. Since $\boldsymbol{E}_l^{Lang}$ and $\boldsymbol{E}_s^{Script}$ provide language and script-specific information, we expect that $\boldsymbol{h}_i$ will be more language-neutral (§5.3), which is beneficial to zero-shot crosslingual transfer (§4.3).

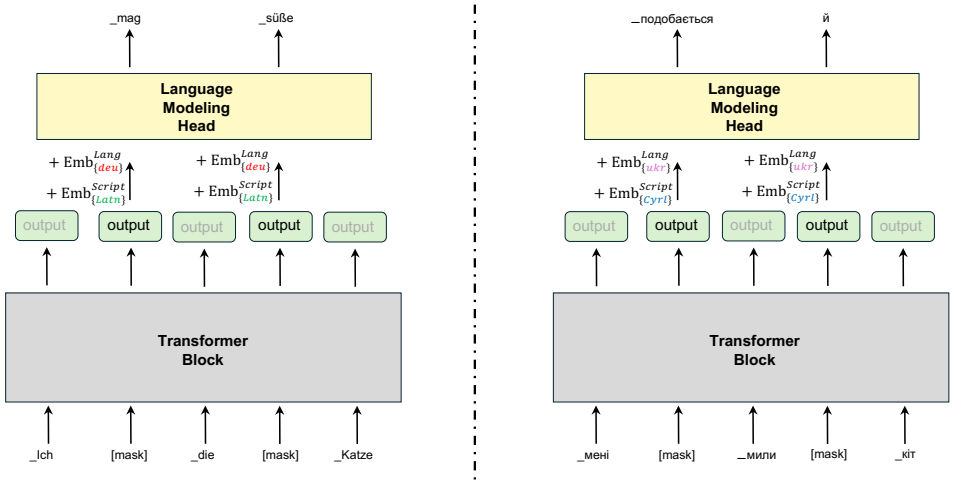


Figure 2: Illustration of LANGSAMP applied to a German sentence (left) and a Ukrainian sentence (right), both meaning "I like the cute cat". Language and script embeddings are added to the outputs from the transformer block. The resulting representation is used to predict the original tokens at the [mask] positions in MLM training.

3.3 Fine-tuning on Downstream tasks

Since we only leverage language and script embedding in the pretraining for MLM, the core architecture (token embeddings + Transformer blocks) remains the same as most mainstream mPLMs, such as XLM-R. In this way, we **do not** need any language or script IDs as input to obtain the Transformer output, i.e., the final contextualized embeddings $\boldsymbol{H}$. This means our pretrained model can be fine-tuned in the standard way in the NLP pipeline. Specifically, for any downstream tasks that require a task-specific classifier (either token-level or sequence-level tasks), we can feed the final contextualized embeddings $\boldsymbol{H} = [\boldsymbol{h}_1, \boldsymbol{h}_2, \cdots, \boldsymbol{h}_n]$ to the classifier and update the model parameters according to the fine-tuning objective, where language or script embeddings are not participating at all. In addition, as $\boldsymbol{H}$ is more language-neutral thanks to LANGSAMP, we expect the representations to boost zero-shot crosslingual transfer (§4.3).

It is important to note that we do not increase the number of parameters used to compute token or sentence representations, as the auxiliary language/script embeddings are employed only during pretraining. This contrasts with prior work, which often introduces additional components – such as Adapters – during downstream fine-tuning (Pfeiffer et al., 2022; Balne et al., 2024). Consequently, any improvements in downstream task performance can only be attributed to enhanced representations learned by the Transformer itself, rather than to added model capacity because of more parameters.

4 Experiments

4.1 Setups

Training Corpora and Tokenizer We use Glot500-c (ImaniGooghari et al., 2023), a corpus that has monolingual data from more than 500 languages written in 30 different scripts. We treat each language-script as a separate entity and refer to those covered by XLM-R (Conneau et al., 2020) as *head languages*, whereas the remaining are *tail languages* (also low-resource languages). We use the tokenizer of Glot500-m (ImaniGooghari et al., 2023), which is a SentencePiece Unigram tokenizer (Kudo and Richardson, 2018; Kudo, 2018) whose vocabulary is merged from the subwords in XLM-R and new subwords learned from Glot500-c.

Continued pretraining We use the weights from XLM-R to initialize our LANGSAMP model for MLM pretraining. **Language and script embeddings are randomly initialized with dimensions $\mathbb{R}^{610\times768}$ and $\mathbb{R}^{30\times768}$ respectively**. We continually train our model on Glot500-c, where we sample data from a multinomial distribution with a temperature of 0.3, to increase the amount of training instances of low- and medium-resource languages. We use AdamW optimizer (Kingma and Ba, 2015; Loshchilov and Hutter, 2019) with $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\epsilon =$ 1e-6. The initial learning rate is set to 5e-5. The effective batch size is 1,024 in each training step, where the gradient accumulation is 8 and the per-GPU batch size is 32. We train the model on 4 NVIDIA RTX6000 GPUs. Each training instance in a batch contains

	tail		head		Latn		non-Latn		all	
	Baseline	LangSAMP	Baseline	LangSAMP	Baseline	LangSAMP	Baseline	LangSAMP	Baseline	LangSAMP
SR-B	36.9 (0.0)	**39.5** (0.0)	60.6 (0.0)	**61.3** (0.0)	40.7 (0.0)	**42.8** (0.0)	51.2 (0.0)	**53.5** (0.0)	42.9 (0.0)	**45.1** (0.0)
SR-T	56.9 (0.0)	**58.6** (0.0)	74.8 (0.0)	**76.1** (0.0)	67.5 (0.0)	**68.7** (0.0)	73.7 (0.0)	**75.6** (0.0)	69.7 (0.0)	**71.1** (0.0)
Taxi1500	47.1 (4.8)	**50.8** (2.4)	59.9 (2.9)	**61.2** (1.2)	48.2 (4.6)	**51.7** (2.1)	58.8 (3.1)	**60.1** (1.7)	50.3 (4.2)	**53.4** (2.0)
SIB200	69.0 (1.4)	**70.2** (1.9)	82.2 (1.4)	**82.6** (1.2)	72.1 (1.3)	**73.1** (1.8)	81.1 (1.5)	**81.7** (1.2)	75.0 (1.3)	**75.9** (1.6)
NER	60.1 (0.6)	**60.8** (0.8)	64.0 (0.6)	**64.1** (0.6)	67.0 (0.5)	**67.6** (0.6)	**53.9** (0.7)	**53.9** (0.5)	62.2 (0.5)	**62.6** (0.6)
POS	61.3 (1.0)	**61.4** (0.9)	76.0 (0.4)	**76.2** (0.4)	**74.6** (0.5)	74.5 (0.4)	66.2 (1.0)	**66.8** (0.8)	71.5 (0.6)	**71.6** (0.5)

Table 1: Performance of LangSAMP and baseline on six downstream tasks across five random seeds. We report the performance by grouping languages according to two characteristics: (1) whether it is a head or a tail language, and (2) whether it is written in Latin script or non-Latin script. The average performance within each group and the standard deviation (in parentheses) are computed. LangSAMP consistently achieves on-par performance or outperforms the baseline across all groups and downstream tasks. **Bold**: best result for each group in each task.

sentences from **the same language-script** which are concatenated to a chunk of 512 tokens. Each batch contains instances from **different language-scripts**. We store checkpoints every 5K steps and apply early stopping with the best average performance on downstream tasks. We set the maximum steps to 150K. The training takes about 4 weeks.

Baseline To validate LangSAMP, we create a baseline where language and script embeddings are not used. This baseline can be regarded as a reproduction of Glot500-m (ImaniGooghari et al., 2023). For a fair comparison, the training hyperparameters and training data (100% data of Glot500-c) are the same as LangSAMP. However, in our ablation study §5.1, due to a constrained computing budget, we cannot continually pretrain model variants on full Glot500-c for validating each component individually (with/without language or script embeddings). Instead, we create such variants and pretrain them using a small portion (5%) of Glot500-c. As a result, the baseline model in Table 1 is different from the vanilla model in Table 2.

4.2 Downstream Tasks

We consider the following three evaluation types, with two datasets for each type. The evaluation is done in an English-centric zero-shot crosslingual transfer style for evaluation types that require fine-tuning. That is, we first fine-tune the pretrained model on the English train set, then select the best checkpoint on the English development set, and finally evaluate the best checkpoint on the test sets of all other languages. For Sentence Retrieval, which does not involve any fine-tuning, we simply use English as the retrieval query language. For all tasks, only a subset of languages (head and tail languages) supported by Glot500-c are considered. We show the detailed information of the used dataset and hyperparameter settings in §A. We introduce the evaluation types and datasets in the following.

Sentence Retrieval. We use Bible (SR-B) and Tatoeba (Artetxe and Schwenk, 2019) (SR-T). The pairwise similarity for retrieving the target sentences is calculated using the mean pooling of contextualized word embeddings at the 8th layer.

Text Classification. We use Taxi1500 (Ma et al., 2023) and SIB200 (Adelani et al., 2024). The former is a dataset based on the Bible, whereas the latter is based on FLORES-200 (Costa-jussà et al., 2022) with more modern genres like technology.

Sequence Labeling. We use WikiANN for named entity recognition (NER) (Pan et al., 2017) and Universal Dependencies (de Marneffe et al., 2021) for Part-Of-Speech (POS) tagging.

4.3 Results and Discussion

We evaluate the LangSAMP model and baseline to understand how the integration of language and script embeddings influences crosslingual transfer. We group the transfer target languages based on two characteristics: (1) whether it is a head or tail language, and (2) whether it is written in Latin or a non-Latin script. This grouping aims to directly identify the effectiveness of LangSAMP on low-resource languages and languages written in a less common script. The results are shown in Table 1.

Both tail and head languages benefit. We observe consistent improvements in tail and head languages across tasks. The enhancement is more obvious in tail languages. For example, LangSAMP improves the performance by 7% for tail languages vs 1% for head languages in SR-B. A similar phenomenon can also be seen for other tasks. This pattern indicates that LangSAMP can be more

	SR-B			SR-T			Taxi1500			SIB200			NER			POS		
	tail	head	all	tail	head	all	tail	head	all	tail	head	all	tail	head	all	tail	head	all
vanilla model	11.9	56.4	23.2	46.0	77.7	68.6	18.1	58.6	28.4	56.1	**83.0**	68.3	55.1	62.8	59.3	49.9	75.7	67.8
w/ $\boldsymbol{E}^{Lang}$	13.1	57.9	24.5	**49.1**	79.0	70.5	18.3	58.5	28.5	57.2	82.7	68.8	**55.2**	**63.0**	**59.5**	49.9	75.8	67.8
w/ $\boldsymbol{E}^{Script}$	12.5	57.4	23.9	48.3	78.4	69.8	18.5	57.0	28.2	56.6	82.1	68.2	55.1	62.4	59.0	**50.8**	**76.2**	**68.4**
w/ $\boldsymbol{E}^{Lang}$ and $\boldsymbol{E}^{Script}$	**13.4**	**58.7**	**24.9**	**49.1**	**79.5**	**70.8**	**20.6**	**58.8**	**30.3**	**57.9**	**83.0**	**69.3**	54.9	61.6	58.6	49.7	75.6	67.6

Table 2: Ablation study. We investigate the effectiveness of language and script embeddings on downstream performance. Note that the vanilla model and w/ $\boldsymbol{E}^{Lang}$ and $\boldsymbol{E}^{Script}$ are different from Baseline and LANGSAMP in Table 1 because of the smaller pretraining data size. By including both types of embeddings, the model achieves the overall best performance among all variants. **Bold** (underlined): best (second-best) result for each column.

helpful for those tail languages, for which the training data is scarce. With the help of language embeddings sharing the burden, the LANGSAMP model can have more language-neutral representations for these languages, resulting in better performance.

Both non-Latin and Latin languages benefit. We observe similar consistent improvements when grouping languages into Latin or non-Latin languages. Different from the trend seen in tail/head groups, we see that no group shows an obvious larger enhancement compared to the other group. This can be explained by the fact that head and tail languages are distributed more equally in Latn and non-Latn groups. In addition, the improvements indicate the incorporation of script embeddings is helpful. By decoupling some script-specific information from the representations, the output generated by the backbone is more script-neutral, leading to better crosslingual transfer across scripts.

Improvements can vary slightly across tasks. We observe more consistent large improvements for sequence-level tasks – retrieval and classification – where LANGSAMP outperforms the baseline in all groups. However, on sequence labeling tasks, LANGSAMP achieves very close performance to the baseline. For example, LANGSAMP scores are 0.1 less compared to the baseline on NER. This could be related to the difficulty of the tasks: both NER and POS are relatively easy tasks, and models can transfer well in prevalent classes, e.g., *nouns*, through shared vocabulary (ImaniGooghari et al., 2023; Liu et al., 2024a). Therefore, decoupling language- or script-specific information from the Transformer output can be less helpful for these tasks. Nevertheless, the overall improvements across tasks indicate the superiority of LANGSAMP compared with the baseline.

5 Analysis

5.1 Ablation Study

In the ablation study, we want to explore the effectiveness of language embeddings and script embeddings individually. However, due to a limited computation budget, we cannot run experiments on the full corpora for each variant. Therefore, we select 5% data for each language from Glot500-c and continually pretrain XLM-R using the same hyperparameters used in the main experiments described in §4.1. Specifically, we consider four variants: **a**) model without language/script embeddings; **b**) model with only language embeddings; **c**) model with only script embeddings; and **d**) model with both language and script embeddings. The performance of each variant is shown in Table 2.

Either language or script embeddings help. The vanilla model achieves the overall worst performance among all model variants. As long as language or script embeddings are included, we generally observe a consistent improvement across all downstream tasks. This indicates that both language and script embeddings can share the burden of encoding too much language- and script-specific information in the token representations. As a result, the representations generated by the model variants with language or script embeddings are more language-neutral, benefiting the crosslingual transfer. The best overall performance is achieved when both language and script embeddings are used, suggesting that decoupling both language- and script-specific information would be the best option for improving crosslingual transfer.

Improvement varies across task types. Similar to the findings in §4.3, we observe that including the auxiliary embeddings is very helpful for sequence-level tasks, especially sentence retrieval, where we observe the highest enhancement, while

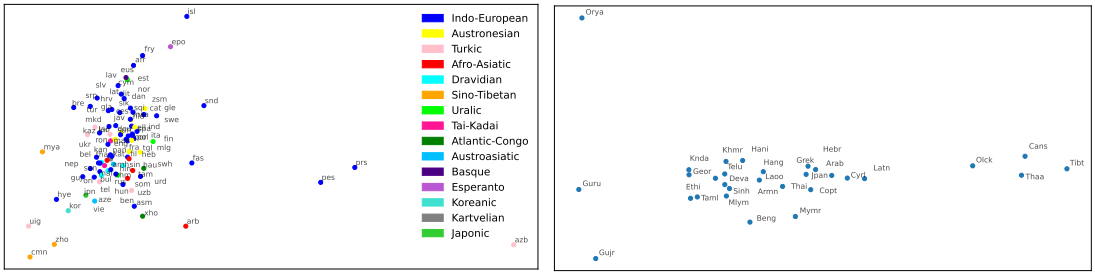


Figure 3: PCA visualizations of head language embeddings (left) and script embeddings (right). We see that some related languages and scripts are close to each other, indicating that they encode language- and script-specific information. Data imbalance may have caused some languages/scripts with limited data to appear as outliers.

less helpful for token-level tasks. It is also noticeable that including language embeddings is the most effective for sentence retrieval (either best or the second best per column). On the other hand, the sequence labeling task does not enjoy large improvements: most model variants achieve on-par performance with each other. The reason has been discussed in §4.3: NER and POS are relatively simple tasks since models can transfer easily in prevalent classes. Nevertheless, the overall results show the effectiveness of the auxiliary embeddings.

5.2 Qualitative Exploration: Visualization

We visualize language and script embeddings in Figure 3. Only head language embeddings are chosen for better readability. We observe that similar or related languages are located close to each other. For example, **cmn** and **zho** (simplified and traditional Chinese, lower left) are closest to each other, as are **pes** (Iranian Persian) and **prs** (Dari). The languages that are mutually influenced by Chinese to a large extent, **jpn**, **kor**, and **vie**, are also close to each other. Most European languages, as well as Indian languages that belong to the Indo-European family, form a rather dense cluster in the middle.

In the plot on the right, most scripts of the Indian subcontinent are found close to each other (**Deva**, **Telu**, **Mlym**, **Taml**, **Knda**, **Sinh**, **Beng**), despite some outliers (e.g., **Gujr** and **Guru**), probably due to the small amount of data that is written in these scripts. **Hani** and scripts of languages that are mutually influenced (**Hang** and **Jpan**) are not far from each other. The same is true for two very related scripts, **Thai** and **Laoo**. In summary, the learnable language and script embeddings can capture language- and script-specific information in the training, which can be helpful for the language-neutrality of the output of transformer blocks.

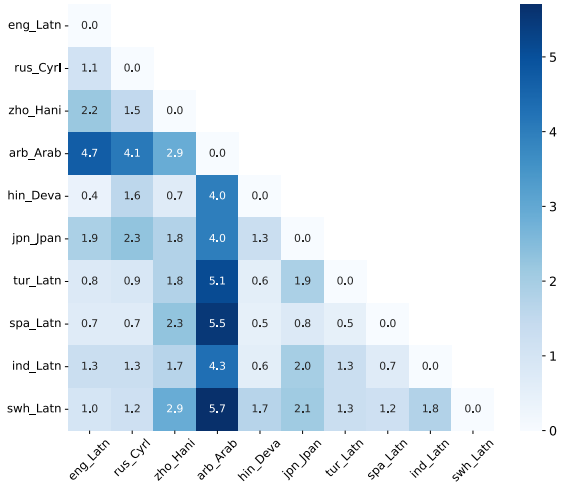


Figure 4: Similarity improvement (by percentage) from baseline to LANGSAMP in terms of the pairwise cosine similarity. Similarity is increased for each pair, indicating better language neutrality of the representations.

5.3 Quantitative Exploration: Similarity

We expect that LANGSAMP can generate more language-neutral representations, meaning that representations of semantically equivalent sentences from different languages are similar. To evaluate this, we selected 10 high-resource languages that differ typologically and use a diverse set of scripts: **eng_Latn**, **rus_Cyrl**, **zho_Hani**, **arb_Arab**, **hin_Deva**, **jpn_Jpan**, **tur_Latn**, **spa_Latn**, **ind_Latn**, and **swa_Latn**. We calculated the pairwise cosine similarity of sentence representations using 100 randomly sampled parallel sentences from SR-B. Sentence representations are obtained by mean-pooling the token representations at the 8th layer, followed by subtracting the language centroid (the average of all 100 sentence representations for that language). We report the pairwise cosine similarity in Figure 5 in §B and show the improvement (by percentage) in Figure 4.

We can observe that the similarity between any

	tail		head		Latn		non-Latn		all	
	English	Donor	English	Donor	English	Donor	English	Donor	English	Donor
Taxi1500	47.3	**48.3**	59.1	**60.3**	48.4	**49.0**	58.1	**60.5**	50.2	**51.2**
SIB200	**67.9**	**67.9**	81.2	**81.6**	71.0	**71.1**	80.3	**80.6**	74.0	**74.2**
NER	61.2	**61.7**	64.1	**65.6**	**67.5**	66.9	54.6	**58.5**	62.8	**63.8**
POS	**63.2**	53.8	**77.0**	72.3	**75.5**	68.4	**68.1**	63.6	**72.8**	66.6

Table 3: Performance of LANGSAMP, using English vs the closest donor language (based on cosine similarity induced from language embeddings) as the source language for zero-shot crosslingual transfer. Each number is the average over all target languages in a class. **Bold**: the result that is better for an English/Donor comparison.

	Taxi1500		SIB200		NER		POS	
tha	eng	jpn	eng	jpn	eng	jpn	eng	jpn
	63.8	**63.8**	85.4	**85.7**	2.1	**10.2**	**58.3**	27.5
yue	eng	zho	eng	zho	eng	zho	eng	zho
	55.4	**67.7**	-	-	25.7	**73.5**	42.6	**80.9**
san	eng	hin	eng	hin	eng	hin	eng	hin
	-	-	72.9	**76.6**	38.4	**53.4**	25.5	**32.7**
urd	eng	hin	eng	hin	eng	hin	eng	hin
	-	-	79.1	**80.6**	65.1	**76.8**	69.7	**89.7**
lin	eng	swh	eng	swh	eng	swh	eng	swh
	47.1	**54.7**	68.2	**73.3**	47.6	**55.9**	-	-
run	eng	swh	eng	swh	eng	swh	eng	swh
	48.0	**55.2**	65.2	**72.7**	-	-	-	-

Table 4: Languages with large improvements when using the closest donor language. In each task, the first/second column indicates results using English/the donor language as the source language. "-" indicates the language is not covered by the task. **Bold**: best result for each language in each task.

two languages is improved in LANGSAMP. The enhancement is especially noticeable for typologically distinct languages using different scripts. For example, arb_Arab is in a different language family and written in a different script compared to the other 9 languages; the similarity involving arb_Arab is greatly improved: 4.7% for eng_Latn and 4.1% rus_Cyrl. Importantly, since LANGSAMP does not incorporate additional parallel data, this improvement is solely attributed to the inclusion of language and script embeddings during pretraining. This indicates that LANGSAMP effectively generates more language-neutral representations by decoupling language- and script-specific features into auxiliary embeddings.

5.4 Case Study: Source Language Selection

Previous studies show language similarities have been useful for selecting good source languages for crosslingual transfer (Lin et al., 2019; Lauscher et al., 2020; Nie et al., 2023; Wang et al., 2023b,a; Lin et al., 2024). We expect this to also apply to the similarities induced by our language embeddings. Therefore, we conduct a case study and use the languages mentioned in §5.3 as the donor languages. When performing the downstream task for a specific target language, instead of always using English as the source language, we select the donor language that is the most cosine-similar to the target language. We evaluate the LANGSAMP model on Taxi1500, SIB200, NER, and POS in a zero-shot crosslingual transfer style. The aggregated results are reported in Table 3, and we select representative target languages that benefit from choosing a good donor language in Table 4.

Effects of donor vary across tasks. Our results suggest that the performance gain from using a donor language varies across tasks. The gain in the text classification task is more consistent than in the sequence labeling task. We assume the primary reason is that the training data for NER and POS are not parallel, and the size of the training data is highly variable across languages. For example, English has much more data than some of the other donor languages for these two tasks.

Non-Latin languages benefit more. For the text classification task, greater improvements can be observed in non-Latin script languages than in Latin script languages. This reflects previous findings that non-Latin script languages are less represented in mPLMs (Muller et al., 2021) and indicates the effectiveness of leveraging language embeddings in selecting better donor languages for them.

Donor is frequently from the same family. We find that language embeddings frequently identify a donor language of the same family as the target language, leading to a large performance improvement over English as the source. For example, as shown in Table 4, **zho_Hani** as a donor language for **yue_Hani** leads to large performance

gains on all three tasks. Similar gains are seen using **hin_Deva** for **san_Deva**. Positive effects can also be found across scripts, as in the case of using **hin_Deva** for **urd_Arab**, two very similar languages written in different scripts.

Interesting cases of unrelated donors. We also notice some interesting cases where the closest donor language is not or only partially related to the target language, but nevertheless aids transfer performance as shown in Table 4. For example, **jpn_Jpan** has a positive effect for **tha_Thai**. Similarly, for **tuk_Latn**, using **rus_Cyrl** as the source achieves better transfer performance than English.

6 Conclusion

We propose LANGSAMP, a multilingual pretraining approach that leverages auxiliary language and script embeddings to facilitate more language-neutral representations by offloading the burden of encoding language- and script-specific information within the Transformer outputs. These embeddings are added to the output of the transformer blocks before being fed into the language modeling head for decoding. In this way, we keep the model structure simple, allowing it to function as a universal text encoder, without requiring language or script IDs as input, while easing the burden of the output encoding too much language- and script-specific information. Through extensive experiments, we show LANGSAMP consistently outperforms the baseline on various downstream tasks, especially in sequence-level tasks. Our ablation study confirms the effectiveness of both language and script embeddings. LANGSAMP exhibits improved language neutrality, as reflected by increased pairwise similarity across all donor languages. Furthermore, our case study demonstrates that the byproducts – auxiliary language/script embeddings – encode language- and script-specific information, which can facilitate the selection of optimal source languages for more effective crosslingual transfer.

7 Future Direction

Looking ahead, a promising research direction is to further explore and refine the use of auxiliary language/script embeddings to guide language-neutral representation learning, particularly in the middle layers of multilingual models. Prior studies have shown that intermediate layers of encoder-only or decoder-only models often exhibit higher language neutrality (Jalili Sabet et al., 2020; ImaniGooghari et al., 2023; Zhao et al., 2024; Li et al., 2025).

Future work could investigate ways to explicitly steer middle-layer representations toward greater neutrality, for example, by combining auxiliary embeddings with layer-specific objectives or contrastive learning alignment techniques.

Furthermore, language and script embeddings hold potential for enhancing controlled multilingual generation in decoder-only and encoder-decoder models, enabling more accurate and consistent generation in the target language based on instructions provided in the prompt.

Limitations

Due to the constraints of computing resources, we are not able to continue pretraining the model using the full Glot500-c data in **our ablation study**. However, as all variants are trained in a strictly controlled environment, their results can be compared in a fair way, and the consistent improvement suggests the effectiveness of the language embeddings.

In addition, we do not consider the possibility of introducing language and script embeddings before the Transformer blocks. Although this is also a possible architecture, it does not fulfill our aim and therefore is not relevant to us. Our primary prerequisite is that the resulting model can work as a universal text encoder without any language or script IDs as input, just like most highly multilingual models (e.g., XLM-R (Conneau et al., 2020) and mBERT (Devlin et al., 2019)). LANGSAMP only requires language or script IDs in the pretraining stage. After that, the backbone (token embeddings + the Transformer blocks) acts exactly as a universal text encoder. Investigating whether architectures that integrate language/script embeddings before the Transformer could improve language representations at scale is outside the scope of our work, but we consider it a promising direction for future research.

Another potential limitation is the coverage of languages and scripts. Our model uses 610 languages and 30 scripts from Glot500-c. For low-resource languages not supported by our model, we can still generate representations since language IDs are not required as input. However, without a corresponding language embedding, it becomes challenging to select the optimal donor language for crosslingual transfer. Nonetheless, when adapting to these languages, the language embeddings

can be expanded, similar to the approach commonly used for vocabulary extension.

Finally, while our results support a strong correlation between improved language neutrality and enhanced crosslingual transfer, we acknowledge that this relationship is not necessarily causal. However, prior studies have shown that improved language neutrality or alignment does not always yield better downstream outcomes (Gaschi et al., 2023; Hua et al., 2024; Liu et al., 2025), suggesting that language neutrality alone may not be a sufficient condition. We believe further research is needed to understand when and how language-neutral representations contribute effectively to transfer, which remains an open and important question for the multilingual NLP community.

Acknowledgments

This work was funded by Deutsche Forschungsgemeinschaft (project SCHU 2246/14-1) and The European Research Council (NonSequeToR, grant #740516).

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A Settings and Hyperparameters

We show the information of the evaluation datasets and used measures in Table 5 and introduce the detailed settings and hyperparameters as follows.

Sentence Retrieval We use English-aligned sentences (up to 500 and 1000 for SR-B and SR-T, respectively) from languages covered by Glot500-c (ImaniGooghari et al., 2023). No fine-tuning is needed for this evaluation type: we directly use each model as a text encoder and generate the sentence-level representation by averaging the contextual token embeddings at the **8th** layer, similar to previous work (Jalili Sabet et al., 2020; ImaniGooghari et al., 2023; Liu et al., 2024a). We perform retrieval by sorting the pairwise similarities.

Text Classification We add a 6-class or 7-class (for Taxi1500 and SIB200, respectively) sequence-level classification head onto the backbone model (no language or script IDs are required as input since the language modeling head is not needed in this sequence-level classification model). By default, we train the model on the English train set and store the best checkpoint on the English validation set. We train all models using AdamW optimizer (Kingma and Ba, 2015; Loshchilov and Hutter, 2019) for a maximum of 40 epochs, with a learning rate of 1e-5 and an effective batch size of 16 (batch size of 8, gradient accumulation of 2). We use a single GTX 1080 Ti GPU for training. The evaluation is done in zero-shot transfer: we directly apply the best checkpoint to the test sets of all other languages.

Sequence Labeling We add a 7-class or 18-class (for NER and POS, respectively) token-level classification head onto the backbone model (no language or script IDs are required as input since the language modeling head is not needed in this token-level classification model). Similarly, we train the model on the English train set and store the best checkpoint on the English validation set by default. We train all models using the AdamW optimizer (Kingma and Ba, 2015; Loshchilov and Hutter, 2019) for a maximum of 10 epochs. The learning rate is set to 2e-5, and the effective batch size is

	|head|	|tail|	|Latn|	|non-Latn|	#class	measure (%)
SR-B	94	275	290	79	-	top-10 Acc.
SR-T	70	28	64	34	-	top-10 Acc.
Taxi1500	89	262	281	70	6	F1 score
SIB200	78	94	117	55	7	F1 score
NER	89	75	104	60	7	F1 score
POS	63	28	57	34	18	F1 score

Table 5: Information of the evaluation datasets and used measures. |head| (resp. |tail|): number of head (resp. tail) language-scripts. |Latn| (resp. |non-Latn|): number of languages written in Latin script (resp. non-Latn scripts). #class: the number of the categories if it belongs to a text classification or sequence labeling task.

set to 32 (batch size of 8, gradient accumulation of 4). The training is done on a single GTX 1080 Ti GPU. The evaluation is done in zero-shot transfer: we directly apply the best checkpoint to the test sets of all other languages.

B Pairwise Cosine Similarity

As introduced in §5.3, we select 10 topologically different languages that are written in diverse scripts to assess the language neutrality: **eng_Latn**, **rus_Cyrl**, **zho_Hani**, **arb_Arab**, **hin_Deva**, **jpn_Jpan**, **tur_Latn**, **spa_Latn**, **ind_Latn**, and **swa_Latn**. We report the pairwise cosine similarity for the baseline and LANGSAMP in Figure 5.

It can be observed that the similarity between any two languages in LANGSAMP is consistently higher than in the baseline. The absolute increase is small in general, due to the fact that (1) without the introduction of the auxiliary language and script embeddings, the baseline already assigns good similarity to translations and (2) LANGSAMP does not introduce any additional parallel data in the pretraining, which is usually regarded as important to improve the similarity. Nevertheless, the consistent improvement indicates that LANGSAMP effectively improves the language neutrality by decoupling language- and script-specific features into auxiliary embeddings.

C Results for Each Language Family

We report the aggregated results for each language family for each task in Table 6. We see consistent improvement for all language families in sentence retrieval and text classification tasks. For sequence tagging tasks, LANGSAMP achieves similar performance compared with the baseline. This trend is similar to the main results we report in §4.3.

D Complete Crosslingual Transfer Results

We report the complete results of English-centric zero-shot crosslingual performance of baseline and LANGSAMP for all tasks and languages in Table 7, 8 (**SR-B**), Table 9 (**SR-T**), Table 10, 11(**Taxi1500**), 12 (**SIB200**), Table 13 (**NER**), and Table 14 (**POS**). Each result is the average over fine-tuning the baseline or LANGSAMP under five random seeds.

E Transfer Results Using English and Closest Donor Language

We report the complete results of the zero-shot crosslingual performance of LANGSAMP when using English and the closest donor language as the source language in Table 15, 16 (**Taxi1500**), 17 (**SIB200**), Table 18 (**NER**), and Table 19 (**POS**). Each result is directly obtained from a single run. We fine-tune the LANGSAMP using different donor languages under the same random seed.

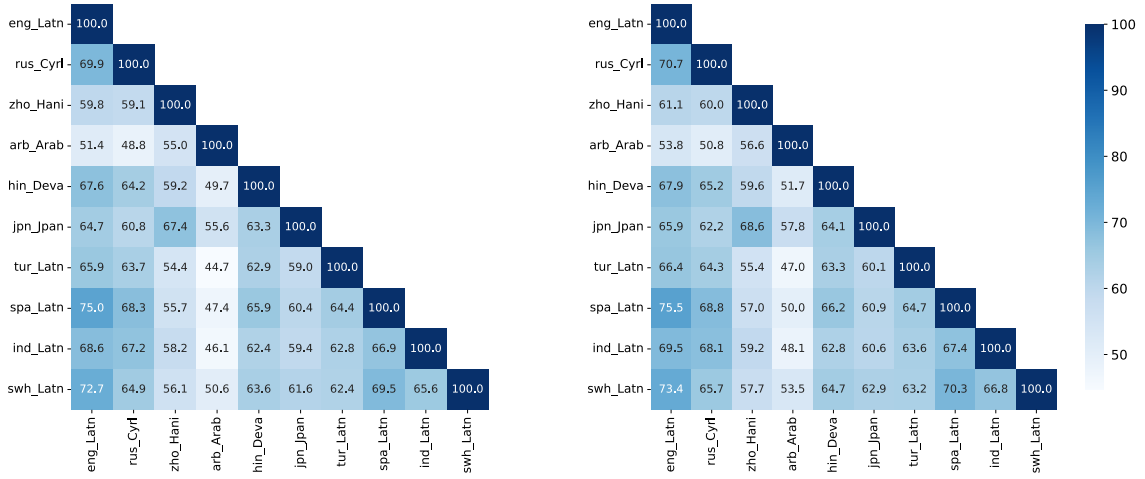


Figure 5: Comparison between baseline (left) and LANGSAMP (right) in terms of the pairwise cosine similarity. LANGSAMP achieves better similarity for each pair, indicating improved language neutrality of the representations.

	(indo1319, 93)	(atla1278, 69)	(aust1307, 55)	(turk1311, 23)	(sino1245, 23)	(maya1287, 15)	(afro1255, 12)	(other, 79)	(all, 369)
SR-B									
Baseline	61.4	37.3	42.9	60.9	31.6	15.5	29.5	31.3	42.9
LANGSAMP	**62.0**	**40.2**	**45.1**	**63.3**	**34.8**	**15.7**	**32.0**	**34.6**	**45.1**
	(indo1319, 54)	(atla1278, 2)	(aust1307, 7)	(turk1311, 7)	(sino1245, 3)	(maya1287, 0)	(afro1255, 5)	(other, 20)	(all, 98)
SR-T									
Baseline	74.2	50.0	48.7	71.3	81.7	-	52.1	68.7	69.7
LANGSAMP	**75.2**	**50.6**	**50.2**	**74.6**	**83.0**	-	**54.2**	**70.5**	**71.1**
	(indo1319, 87)	(atla1278, 68)	(aust1307, 51)	(turk1311, 18)	(sino1245, 22)	(maya1287, 15)	(afro1255, 11)	(other, 79)	(all, 351)
Taxi1500									
Baseline	60.8	42.6	51.6	60.3	49.5	42.4	35.7	46.1	50.3
LANGSAMP	**62.6**	**46.8**	**55.2**	**62.7**	**53.6**	**45.5**	**38.8**	**49.1**	**53.4**
	(indo1319, 71)	(atla1278, 33)	(aust1307, 17)	(turk1311, 10)	(sino1245, 5)	(maya1287, 0)	(afro1255, 13)	(other, 23)	(all, 172)
SIB200									
Baseline	82.1	59.0	76.4	80.5	67.4	-	73.0	75.1	75.0
LANGSAMP	**82.7**	**60.5**	**78.0**	**81.8**	**68.7**	-	**73.1**	**75.7**	**75.9**
	(indo1319, 94)	(atla1278, 5)	(aust1307, 12)	(turk1311, 12)	(sino1245, 7)	(maya1287, 0)	(afro1255, 6)	(other, 28)	(all, 164)
NER									
Baseline	66.8	60.4	58.8	**62.1**	**37.1**	-	53.8	56.7	62.2
LANGSAMP	**67.2**	**61.2**	**59.5**	61.3	36.6	-	**55.5**	**57.2**	**62.6**
	(indo1319, 54)	(atla1278, 2)	(aust1307, 4)	(turk1311, 5)	(sino1245, 3)	(maya1287, 1)	(afro1255, 6)	(other, 16)	(all, 91)
POS									
Baseline	**78.0**	**61.9**	**74.4**	**72.1**	33.3	**61.1**	64.1	60.3	71.5
LANGSAMP	**78.0**	61.0	**74.4**	71.9	**35.8**	58.8	**64.9**	**60.6**	**71.6**

Table 6: Aggregated performance of the baseline and LANGSAMP for 7 major language families on all tasks. We report the average performance for **indo1319** (Indo-European), **atla1278** (Atlantic-Congo), **aust1307** (Austronesian), **turk1311** (Turkic), **sino1245** (Sino-Tibetan), **maya1287** (Mayan), and **afro1255** (Afro-Asiatic). We classify the remaining languages into the group "**other**". In addition, we report the average over all languages (group "**all**"). The number of languages in that family is shown in parentheses. **Bold**: best result for each task.

Language	Baseline	LangSAMP	Language	Baseline	LangSAMP	Language	Baseline	LangSAMP	Language	Baseline	LangSAMP
ace_Latn	43.8	**49.4**	ach_Latn	37.6	**40.6**	acr_Latn	17.6	**18.6**	afr_Latn	**74.2**	72.4
agw_Latn	31.0	**38.2**	ahk_Latn	3.4	**3.8**	aka_Latn	41.8	**48.4**	aln_Latn	**70.0**	**70.0**
als_Latn	**54.4**	**54.4**	alt_Cyrl	53.8	**57.0**	alz_Latn	36.2	**37.4**	amh_Ethi	44.4	**51.2**
aoj_Latn	15.6	**18.6**	arb_Arab	9.6	**11.6**	arn_Latn	18.2	**23.0**	ary_Arab	11.2	**13.0**
arz_Arab	**15.2**	**15.2**	asm_Beng	**59.2**	59.0	ayr_Latn	37.6	**46.0**	azb_Arab	55.6	**59.0**
aze_Latn	73.4	**75.4**	bak_Cyrl	58.8	**62.2**	bam_Latn	38.4	**44.8**	ban_Latn	33.0	**33.2**
bar_Latn	32.2	**34.0**	bba_Latn	26.2	**31.0**	bbc_Latn	**60.8**	58.8	bci_Latn	**12.0**	11.8
bcl_Latn	75.4	**79.0**	bel_Cyrl	**70.6**	69.6	bem_Latn	51.0	**54.4**	ben_Beng	53.4	**55.4**
bhw_Latn	28.4	**30.6**	bim_Latn	31.4	**42.8**	bis_Latn	45.2	**50.8**	bod_Tibt	29.6	**33.6**
bqc_Latn	27.4	**29.2**	bre_Latn	**31.8**	30.0	bts_Latn	**62.4**	62.0	btx_Latn	**57.2**	55.8
bul_Cyrl	79.8	**80.0**	bum_Latn	32.8	**35.2**	bzj_Latn	69.8	**70.2**	cab_Latn	11.6	**11.8**
cac_Latn	10.8	**11.8**	cak_Latn	**17.8**	16.6	caq_Latn	26.0	**29.8**	cat_Latn	**85.4**	83.2
cbk_Latn	54.8	**56.2**	cce_Latn	41.8	**45.4**	ceb_Latn	70.4	**70.6**	ces_Latn	**68.2**	67.0
cfm_Latn	34.4	**38.8**	che_Cyrl	10.2	**11.2**	chk_Latn	35.2	**43.0**	chv_Cyrl	45.0	**54.4**
ckb_Arab	31.2	**32.8**	cmn_Hani	**41.4**	40.8	cnh_Latn	38.2	**43.2**	crh_Cyrl	67.2	**70.0**
crs_Latn	**85.6**	84.4	csy_Latn	40.2	**49.6**	ctd_Latn	44.4	**50.6**	ctu_Latn	**16.6**	16.0
cuk_Latn	**17.0**	**17.0**	cym_Latn	**45.6**	43.8	dan_Latn	**72.4**	71.8	deu_Latn	73.8	**74.0**
djk_Latn	**38.0**	**38.0**	dln_Latn	46.6	**51.4**	dtp_Latn	17.0	**17.8**	dyu_Latn	33.0	**40.2**
dzo_Tibt	28.4	**33.0**	efi_Latn	41.6	**53.6**	ell_Grek	48.2	**49.2**	enm_Latn	**69.4**	**69.4**
epo_Latn	**67.4**	65.8	est_Latn	**66.4**	66.0	eus_Latn	23.8	**24.2**	ewe_Latn	33.2	**34.8**
fao_Latn	**79.8**	78.4	fas_Arab	80.2	**84.2**	fij_Latn	30.0	**31.0**	fil_Latn	**77.6**	77.2
fin_Latn	65.4	**66.0**	fon_Latn	20.2	**25.2**	fra_Latn	**87.4**	87.2	fry_Latn	**47.0**	44.0
gaa_Latn	34.4	**40.6**	gil_Latn	30.0	**31.6**	giz_Latn	32.4	**36.4**	gkn_Latn	20.4	**24.2**
gkp_Latn	13.2	**14.6**	gla_Latn	**39.0**	38.0	gle_Latn	**41.2**	38.4	glv_Latn	37.2	**38.6**
gom_Latn	33.2	**36.0**	gor_Latn	21.8	**23.0**	grc_Grek	44.4	**47.0**	guc_Latn	**9.8**	8.2
gug_Latn	28.2	**31.2**	guj_Gujr	**69.8**	67.6	gur_Latn	17.6	**18.2**	guw_Latn	36.8	**45.4**
gya_Latn	27.6	**32.6**	gym_Latn	**13.6**	13.0	hat_Latn	**76.4**	74.6	hau_Latn	57.6	**59.6**
haw_Latn	28.0	**30.4**	heb_Hebr	21.6	**23.0**	hif_Latn	33.2	**34.6**	hil_Latn	74.0	**79.8**
hin_Deva	**75.6**	74.6	hin_Latn	34.2	**36.2**	hmo_Latn	44.2	**57.0**	hne_Deva	71.6	**73.6**
hnj_Latn	39.6	**46.6**	hra_Latn	43.4	**46.4**	hrv_Latn	**80.4**	79.8	hui_Latn	19.8	**22.0**
hun_Latn	65.6	**69.0**	hus_Latn	14.8	**16.2**	hye_Armn	62.8	**65.6**	iba_Latn	70.2	**71.6**
ibo_Latn	**32.4**	31.6	ifa_Latn	26.2	**29.0**	ifb_Latn	**28.6**	**28.6**	ikk_Latn	30.2	**46.4**
ilo_Latn	53.4	**54.4**	ind_Latn	78.4	**78.6**	isl_Latn	71.0	**71.8**	ita_Latn	76.2	**76.8**
ium_Latn	20.0	**23.2**	ixl_Latn	13.8	**14.4**	izz_Latn	19.6	**22.6**	jam_Latn	**61.0**	59.2
jav_Latn	**55.4**	52.0	jpn_Jpan	65.8	**67.6**	kaa_Cyrl	71.2	**75.0**	kaa_Latn	32.0	**37.6**
kab_Latn	12.2	**13.4**	kac_Latn	22.2	**27.0**	kal_Latn	12.6	**16.8**	kan_Knda	50.0	**52.8**
kat_Geor	49.6	**52.4**	kaz_Cyrl	69.4	**70.4**	kbp_Latn	21.8	**26.8**	kek_Latn	16.6	**18.6**
khm_Khmr	39.4	**43.0**	kia_Latn	24.6	**28.8**	kik_Latn	44.4	**48.4**	kin_Latn	56.6	**60.2**
kir_Cyrl	69.8	**70.2**	kjb_Latn	23.4	**26.0**	kjh_Cyrl	45.6	**50.6**	kmm_Latn	33.8	**38.0**
kmr_Cyrl	**42.0**	40.2	kmr_Latn	60.2	**60.4**	knv_Latn	7.0	**8.4**	kor_Hang	60.8	**64.0**
kpg_Latn	42.6	**48.8**	krc_Cyrl	59.8	**62.2**	kri_Latn	61.4	**62.6**	ksd_Latn	31.4	**41.0**
kss_Latn	5.2	**6.0**	ksw_Mymr	26.2	**28.0**	kua_Latn	43.0	**43.8**	lam_Latn	20.4	**22.8**
lao_Laoo	41.6	**47.2**	lat_Latn	56.6	**58.0**	lav_Latn	69.8	**71.2**	ldi_Latn	**22.4**	22.0
leh_Latn	**46.8**	45.8	lhu_Latn	**4.4**	4.2	lin_Latn	64.6	**71.0**	lit_Latn	**67.0**	66.6
loz_Latn	**46.8**	45.6	ltz_Latn	**63.8**	63.2	lug_Latn	37.2	**40.8**	luo_Latn	**42.8**	42.6
lus_Latn	46.6	**53.2**	lzh_Hani	59.8	**62.4**	mad_Latn	42.6	**44.6**	mah_Latn	30.4	**33.8**
mai_Deva	52.6	**56.0**	mal_Mlym	51.6	**57.4**	mam_Latn	**10.2**	**10.2**	mar_Deva	68.4	**71.4**
mau_Latn	2.8	**3.4**	mbb_Latn	22.0	**29.8**	mck_Latn	**55.6**	53.4	mcn_Latn	34.2	**40.8**
mco_Latn	**6.6**	6.4	mdy_Ethi	21.4	**30.6**	meu_Latn	48.8	**52.0**	mfe_Latn	**77.4**	**77.4**

Table 7: Top-10 accuracy of models on **SR-B** (Part I).

Language	Baseline	LANGSAMP	Language	Baseline	LANGSAMP	Language	Baseline	LANGSAMP	Language	Baseline	LANGSAMP
mgh_Latn	17.4	**20.8**	mgr_Latn	**48.6**	47.2	mhr_Cyrl	37.4	**43.2**	min_Latn	**32.4**	29.6
miq_Latn	28.8	**36.8**	mkd_Cyrl	78.4	**78.8**	mlg_Latn	60.2	**61.2**	mlt_Latn	48.0	**50.4**
mos_Latn	32.2	**32.8**	mps_Latn	16.4	**20.6**	mri_Latn	45.6	**55.0**	mrw_Latn	34.0	**40.6**
msa_Latn	43.6	**44.2**	mwm_Latn	24.0	**25.6**	mxv_Latn	**7.0**	**7.0**	mya_Mymr	25.8	**28.0**
myv_Cyrl	26.6	**30.6**	mzh_Latn	24.6	**25.4**	nan_Latn	13.2	**13.6**	naq_Latn	16.8	**26.8**
nav_Latn	**8.6**	**8.6**	nbl_Latn	**49.4**	48.4	nch_Latn	**21.6**	**21.6**	ncj_Latn	18.8	**19.4**
ndc_Latn	32.4	**36.2**	nde_Latn	51.0	**54.8**	ndo_Latn	41.0	**44.0**	nds_Latn	**38.4**	**38.4**
nep_Deva	56.4	**59.0**	ngu_Latn	**26.2**	26.0	nia_Latn	25.6	**28.0**	nld_Latn	**78.4**	78.0
nmf_Latn	25.6	**28.2**	nnb_Latn	33.2	**38.8**	nno_Latn	**76.8**	75.8	nob_Latn	**85.4**	85.0
nor_Latn	**85.8**	83.4	npi_Deva	77.4	**80.8**	nse_Latn	48.4	**51.8**	nso_Latn	46.2	**50.2**
nya_Latn	**57.6**	**57.6**	nyn_Latn	**48.8**	47.4	nyy_Latn	23.4	**24.6**	nzi_Latn	29.2	**34.4**
ori_Orya	51.2	**53.4**	ory_Orya	46.4	**49.8**	oss_Cyrl	41.4	**56.4**	ote_Latn	12.0	**13.2**
pag_Latn	**55.2**	52.2	pam_Latn	37.4	**41.2**	pan_Guru	**46.2**	45.4	pap_Latn	72.8	**75.0**
pau_Latn	17.0	**23.4**	pcm_Latn	**69.8**	69.4	pdt_Latn	**69.4**	66.0	pes_Arab	74.2	**75.2**
pis_Latn	51.4	**54.8**	pls_Latn	27.0	**31.8**	plt_Latn	60.2	**60.8**	poh_Latn	10.6	**11.4**
pol_Latn	73.8	**75.6**	pon_Latn	21.4	**24.0**	por_Latn	**81.8**	81.0	prk_Latn	42.0	**47.4**
prs_Arab	84.6	**87.0**	pxm_Latn	18.2	**19.8**	qub_Latn	30.6	**35.6**	quc_Latn	**18.6**	17.4
qug_Latn	53.6	**59.2**	quh_Latn	40.2	**43.8**	quw_Latn	46.2	**50.4**	quy_Latn	47.4	**54.4**
quz_Latn	59.4	**63.6**	qvi_Latn	49.2	**57.6**	rap_Latn	17.0	**17.8**	rar_Latn	**20.4**	19.8
rmy_Latn	30.4	**32.2**	ron_Latn	**69.4**	69.0	rop_Latn	35.8	**41.4**	rug_Latn	37.8	**38.4**
run_Latn	48.2	**52.4**	rus_Cyrl	74.6	**76.4**	sag_Latn	39.6	**45.4**	sah_Cyrl	43.4	**45.8**
san_Deva	**24.2**	23.6	san_Latn	**7.8**	7.4	sba_Latn	28.0	**29.2**	seh_Latn	67.4	**69.4**
sin_Sinh	45.6	**49.0**	slk_Latn	**69.8**	69.2	slv_Latn	**61.2**	60.8	sme_Latn	35.0	**37.6**
smo_Latn	27.6	**28.8**	sna_Latn	38.4	**41.2**	snd_Arab	**67.2**	65.0	som_Latn	**35.0**	34.8
sop_Latn	**32.4**	28.8	sot_Latn	48.4	**52.4**	spa_Latn	80.8	**81.4**	sqi_Latn	62.2	**64.8**
srm_Latn	**28.2**	26.6	srn_Latn	75.4	**75.6**	srp_Cyrl	**87.2**	85.8	srp_Latn	**85.8**	85.4
ssw_Latn	42.8	**47.0**	sun_Latn	52.0	**54.0**	suz_Deva	21.0	**22.6**	swe_Latn	**78.6**	77.0
swh_Latn	**71.6**	71.4	sxn_Latn	20.6	**20.8**	tam_Taml	47.0	**50.6**	tat_Cyrl	68.2	**70.4**
tbz_Latn	13.2	**18.2**	tca_Latn	10.0	**13.8**	tdt_Latn	50.0	**53.6**	tel_Telu	48.0	**50.2**
teo_Latn	19.4	**19.6**	tgk_Cyrl	69.2	**69.4**	tgl_Latn	**79.6**	78.0	tha_Thai	33.8	**38.0**
tih_Latn	42.2	**46.4**	tir_Ethi	32.2	**34.8**	tlh_Latn	62.0	**66.4**	tob_Latn	**11.6**	11.4
toh_Latn	36.8	**41.8**	toi_Latn	**39.4**	**39.4**	toj_Latn	**14.8**	12.6	ton_Latn	16.0	**16.6**
top_Latn	**6.6**	6.0	tpi_Latn	58.0	**62.2**	tpm_Latn	**27.4**	23.0	tsn_Latn	32.6	**34.6**
tso_Latn	50.0	**51.0**	tsz_Latn	21.2	**25.8**	tuc_Latn	25.6	**32.4**	tui_Latn	29.8	**31.0**
tuk_Cyrl	67.4	**69.4**	tuk_Latn	67.6	**70.0**	tum_Latn	**58.4**	57.0	tur_Latn	70.2	**70.4**
twi_Latn	35.0	**42.0**	tyv_Cyrl	**44.2**	43.4	tzh_Latn	19.0	**19.8**	tzo_Latn	**14.2**	13.6
udm_Cyrl	41.6	**45.2**	uig_Arab	47.4	**50.8**	uig_Latn	57.2	**58.8**	ukr_Cyrl	67.0	**68.0**
urd_Arab	60.4	**61.4**	uzb_Cyrl	80.6	**81.2**	uzb_Latn	**70.0**	68.2	uzn_Cyrl	82.4	**83.0**
ven_Latn	37.2	**42.0**	vie_Latn	68.0	**69.4**	wal_Latn	35.0	**43.4**	war_Latn	42.6	**44.0**
wbm_Latn	37.6	**46.2**	wol_Latn	31.8	**33.2**	xav_Latn	3.8	**4.0**	xho_Latn	42.6	**44.2**
yan_Latn	16.4	**27.2**	yao_Latn	37.4	**37.6**	yap_Latn	15.8	**19.6**	yom_Latn	37.6	**40.0**
yor_Latn	27.4	**28.8**	yua_Latn	**13.2**	12.8	yue_Hani	**17.2**	**17.2**	zai_Latn	29.0	**30.6**
zho_Hani	41.6	**41.8**	zlm_Latn	**84.8**	**84.8**	zom_Latn	39.6	**45.0**	zsm_Latn	90.0	**91.0**
zul_Latn	**51.4**	51.0									

Table 8: Top-10 accuracy of models on **SR-B** (Part II).

Language	Baseline	LangSAMP	Language	Baseline	LangSAMP	Language	Baseline	LangSAMP	Language	Baseline	LangSAMP
afr_Latn	77.9	**80.4**	amh_Ethi	47.0	**52.4**	ara_Arab	**69.4**	68.7	arz_Arab	61.8	**63.9**
ast_Latn	80.3	**84.3**	aze_Latn	82.6	**84.1**	bel_Cyrl	**83.6**	83.0	ben_Beng	72.1	**74.9**
bos_Latn	90.1	**90.4**	bre_Latn	17.4	**18.2**	bul_Cyrl	87.5	**89.2**	cat_Latn	78.2	**78.6**
cbk_Latn	**49.4**	48.0	ceb_Latn	39.0	**42.5**	ces_Latn	**75.7**	73.5	cmn_Hani	87.1	**87.4**
csb_Latn	38.3	**38.7**	cym_Latn	52.2	**55.0**	dan_Latn	91.7	**92.9**	deu_Latn	95.5	**95.7**
dtp_Latn	17.0	**19.3**	ell_Grek	79.3	**82.7**	epo_Latn	71.8	**74.8**	est_Latn	68.2	**69.9**
eus_Latn	52.2	**55.4**	fao_Latn	**77.1**	75.6	fin_Latn	72.3	**74.2**	fra_Latn	**85.3**	85.2
fry_Latn	75.1	**79.2**	gla_Latn	38.4	**38.6**	gle_Latn	44.8	**48.3**	glg_Latn	**77.1**	76.4
gsw_Latn	58.1	**63.2**	heb_Hebr	71.4	**74.9**	hin_Deva	**88.1**	87.3	hrv_Latn	**87.9**	87.5
hsb_Latn	**49.7**	**49.7**	hun_Latn	71.5	**73.2**	hye_Armn	79.1	**81.3**	ido_Latn	54.6	**55.8**
ile_Latn	71.2	**71.5**	ina_Latn	89.2	**90.7**	ind_Latn	88.1	**88.9**	isl_Latn	84.0	**84.5**
ita_Latn	84.1	**85.7**	jpn_Jpan	**77.2**	77.1	kab_Latn	10.8	**11.0**	kat_Geor	71.2	**72.4**
kaz_Cyrl	74.6	**77.7**	khm_Khmr	57.5	**63.0**	kor_Hang	80.8	**81.1**	kur_Latn	49.8	**52.4**
lat_Latn	39.2	**42.1**	lfn_Latn	55.8	**56.8**	lit_Latn	70.4	**72.9**	lvs_Latn	76.2	**78.1**
mal_Mlym	87.5	**91.6**	mar_Deva	79.8	**81.6**	mhr_Cyrl	27.7	**33.4**	mkd_Cyrl	**79.6**	79.4
mon_Cyrl	78.2	**80.5**	nds_Latn	71.3	**72.5**	nld_Latn	92.4	**93.4**	nno_Latn	85.5	**87.4**
nob_Latn	94.5	**95.3**	oci_Latn	**46.6**	44.9	pam_Latn	**10.2**	**10.2**	pes_Arab	86.7	**86.9**
pms_Latn	49.5	**50.9**	pol_Latn	**84.3**	83.4	por_Latn	90.2	**90.7**	ron_Latn	86.0	**86.9**
rus_Cyrl	91.6	**92.1**	slk_Latn	77.9	**78.2**	slv_Latn	**76.2**	75.9	spa_Latn	**88.6**	88.3
sqi_Latn	84.1	**85.2**	srp_Latn	**89.7**	89.6	swe_Latn	89.4	**89.6**	swh_Latn	**45.1**	44.9
tam_Taml	**50.2**	45.0	tat_Cyrl	71.2	**74.6**	tel_Telu	72.6	**74.8**	tgl_Latn	73.9	**74.2**
tha_Thai	75.4	**79.2**	tuk_Latn	62.1	**68.0**	tur_Latn	79.1	**82.0**	uig_Arab	64.7	**68.4**
ukr_Cyrl	84.9	**86.5**	urd_Arab	78.5	**81.7**	uzb_Cyrl	65.0	**67.3**	vie_Latn	**88.9**	88.8
war_Latn	22.7	**25.2**	wuu_Hani	79.0	**82.4**	xho_Latn	54.9	**56.3**	yid_Hebr	65.8	**67.6**
yue_Hani	79.0	**79.3**	zsm_Latn	90.2	**91.0**						

Table 9: Top-10 accuracy of models on **SR-T**.

Language	Baseline	LANGSAMP	Language	Baseline	LANGSAMP	Language	Baseline	LANGSAMP	Language	Baseline	LANGSAMP
ace_Latn	**66.6**	64.5	ach_Latn	36.6	**40.5**	acr_Latn	45.6	**51.4**	afr_Latn	**60.7**	58.9
agw_Latn	53.6	**55.8**	ahk_Latn	**7.6**	7.3	aka_Latn	43.3	**47.4**	aln_Latn	56.0	**57.4**
als_Latn	56.6	**57.1**	alt_Cyrl	48.5	**49.9**	alz_Latn	31.9	**39.1**	amh_Ethi	**8.7**	7.9
aoj_Latn	36.8	**42.5**	arn_Latn	41.9	**45.1**	ary_Arab	33.8	**34.3**	arz_Arab	36.2	**40.5**
asm_Beng	**63.2**	62.4	ayr_Latn	55.9	**57.3**	azb_Arab	**63.7**	62.8	aze_Latn	67.3	**70.3**
bak_Cyrl	**60.8**	58.9	bam_Latn	43.8	**49.6**	ban_Latn	42.6	**47.7**	bar_Latn	46.2	**49.7**
bba_Latn	40.9	**43.3**	bci_Latn	31.3	**33.7**	bcl_Latn	55.0	**61.7**	bel_Cyrl	**59.8**	**59.8**
bem_Latn	45.5	**50.6**	ben_Beng	62.4	**65.9**	bhw_Latn	45.3	**53.2**	bim_Latn	49.4	**49.8**
bis_Latn	67.2	**71.8**	bqc_Latn	32.3	**36.7**	bre_Latn	37.3	**44.0**	btx_Latn	54.0	**64.6**
bul_Cyrl	**65.3**	**65.3**	bum_Latn	39.5	**45.3**	bzj_Latn	66.2	**67.7**	cab_Latn	24.5	**31.0**
cac_Latn	44.8	**45.9**	cak_Latn	53.2	**54.4**	caq_Latn	39.9	**45.7**	cat_Latn	**63.6**	61.6
cbk_Latn	62.2	**68.4**	cce_Latn	42.5	**48.3**	ceb_Latn	53.1	**56.0**	ces_Latn	61.6	**66.3**
cfm_Latn	55.3	**64.8**	che_Cyrl	17.6	**23.2**	chv_Cyrl	56.7	**61.7**	cmn_Hani	67.8	**69.2**
cnh_Latn	61.9	**64.5**	crh_Cyrl	61.2	**64.3**	crs_Latn	**65.9**	64.3	csy_Latn	53.6	**62.8**
ctd_Latn	53.5	**58.5**	ctu_Latn	**52.2**	51.8	cuk_Latn	40.2	**43.3**	cym_Latn	**50.1**	49.4
dan_Latn	62.4	**63.6**	deu_Latn	53.4	**56.7**	djk_Latn	46.5	**54.7**	dln_Latn	49.3	**61.4**
dtp_Latn	50.2	**51.8**	dyu_Latn	48.0	**57.1**	dzo_Tibt	55.9	**57.8**	efi_Latn	52.4	**56.2**
ell_Grek	59.8	**61.3**	eng_Latn	74.3	**75.3**	enm_Latn	**72.2**	70.7	epo_Latn	57.6	**58.8**
est_Latn	56.8	**57.5**	eus_Latn	23.3	**27.8**	ewe_Latn	43.6	**52.4**	fao_Latn	57.5	**59.5**
fas_Arab	**71.7**	70.3	fij_Latn	44.2	**48.5**	fil_Latn	57.5	**58.8**	fin_Latn	58.3	**59.2**
fon_Latn	42.9	**44.0**	fra_Latn	65.4	**70.4**	fry_Latn	40.0	**43.1**	gaa_Latn	40.2	**41.8**
gil_Latn	42.0	**44.5**	giz_Latn	45.1	**49.7**	gkn_Latn	38.3	**43.7**	gkp_Latn	32.1	**37.5**
gla_Latn	48.3	**49.4**	gle_Latn	42.9	**44.9**	glv_Latn	39.7	**43.5**	gom_Latn	35.5	**38.2**
gor_Latn	42.5	**50.8**	guc_Latn	33.4	**38.4**	gug_Latn	36.2	**41.2**	guj_Gujr	68.8	**70.3**
gur_Latn	34.1	**43.3**	guw_Latn	48.4	**52.1**	gya_Latn	**40.6**	39.9	gym_Latn	41.1	**47.2**
hat_Latn	62.8	**65.2**	hau_Latn	54.2	**58.6**	haw_Latn	30.2	**38.3**	heb_Hebr	18.7	**21.7**
hif_Latn	46.2	**47.9**	hil_Latn	65.1	**67.3**	hin_Deva	66.2	**69.0**	hmo_Latn	60.7	**65.3**
hne_Deva	66.3	**67.4**	hnj_Latn	63.4	**66.3**	hra_Latn	51.5	**56.0**	hrv_Latn	63.4	**67.3**
hui_Latn	46.5	**50.2**	hun_Latn	64.2	**67.5**	hus_Latn	38.0	**42.3**	hye_Armn	70.0	**70.9**
iba_Latn	57.6	**61.3**	ibo_Latn	**58.2**	56.8	ifa_Latn	49.1	**55.0**	ifb_Latn	50.3	**50.7**
ikk_Latn	47.8	**51.8**	ilo_Latn	53.2	**60.5**	ind_Latn	76.4	**78.0**	isl_Latn	50.4	**59.3**
ita_Latn	64.8	**66.3**	ium_Latn	57.5	**58.9**	ixl_Latn	32.2	**38.3**	izz_Latn	42.4	**49.4**
jam_Latn	64.2	**68.5**	jav_Latn	45.9	**50.5**	jpn_Jpan	**64.9**	63.1	kaa_Cyrl	59.3	**66.8**
kab_Latn	23.0	**29.9**	kac_Latn	**49.0**	45.6	kal_Latn	32.1	**37.2**	kan_Knda	**67.1**	65.2
kat_Geor	**60.0**	57.2	kaz_Cyrl	**65.2**	62.8	kbp_Latn	35.3	**38.0**	kek_Latn	45.5	**47.4**
khm_Khmr	**69.0**	66.5	kia_Latn	41.3	**52.7**	kik_Latn	42.7	**46.4**	kin_Latn	44.7	**56.7**
kir_Cyrl	67.4	**67.9**	kjb_Latn	46.8	**48.5**	kjh_Cyrl	50.9	**55.8**	kmm_Latn	46.2	**57.4**
kmr_Cyrl	49.9	**51.7**	knv_Latn	44.5	**45.8**	kor_Hang	70.2	**71.6**	kpg_Latn	64.4	**65.4**
krc_Cyrl	57.6	**61.8**	kri_Latn	59.4	**62.8**	ksd_Latn	**54.8**	52.9	kss_Latn	**23.7**	18.9
ksw_Mymr	49.0	**50.8**	kua_Latn	42.6	**45.4**	lam_Latn	33.3	**38.1**	lao_Laoo	**72.7**	70.7
lat_Latn	58.5	**63.0**	lav_Latn	62.8	**63.8**	ldi_Latn	27.6	**35.8**	leh_Latn	45.1	**46.9**
lhu_Latn	24.0	**25.9**	lin_Latn	47.6	**54.7**	lit_Latn	61.2	**61.7**	loz_Latn	51.2	**52.6**
ltz_Latn	**53.3**	51.9	lug_Latn	44.0	**52.6**	luo_Latn	38.7	**43.5**	lus_Latn	48.1	**53.9**
lzh_Hani	61.5	**67.2**	mad_Latn	60.6	**62.5**	mah_Latn	34.3	**45.3**	mai_Deva	**65.1**	64.8
mal_Mlym	**7.2**	6.1	mam_Latn	29.2	**34.9**	mar_Deva	62.4	**62.6**	mau_Latn	**7.0**	5.7
mbb_Latn	52.0	**54.4**	mck_Latn	41.1	**46.0**	mcn_Latn	36.6	**42.8**	mco_Latn	24.4	**26.8**
mdy_Ethi	49.1	**54.4**	meu_Latn	49.4	**57.8**	mfe_Latn	**68.8**	68.5	mgh_Latn	32.9	**34.8**
mgr_Latn	47.3	**50.9**	mhr_Cyrl	**41.6**	40.9	min_Latn	51.2	**53.2**	miq_Latn	52.1	**52.7**
mkd_Cyrl	68.8	**71.7**	mlg_Latn	48.3	**51.9**	mlt_Latn	51.3	**53.2**	mos_Latn	36.4	**44.4**

Table 10: F1 scores of models on **Taxi1500** (Part I).

Language	Baseline	LangSAMP	Language	Baseline	LangSAMP	Language	Baseline	LangSAMP	Language	Baseline	LangSAMP
mps_Latn	51.6	**56.2**	mri_Latn	43.2	**49.8**	mrw_Latn	**48.7**	47.9	msa_Latn	46.7	**48.9**
mwm_Latn	52.5	**58.5**	mxv_Latn	16.0	**27.6**	mya_Mymr	57.2	**57.8**	myv_Cyrl	42.9	**48.1**
mzh_Latn	39.2	**41.9**	nan_Latn	26.2	**33.7**	naq_Latn	40.3	**45.5**	nav_Latn	22.2	**25.7**
nbl_Latn	46.7	**52.9**	nch_Latn	41.8	**46.0**	ncj_Latn	36.2	**42.2**	ndc_Latn	39.3	**44.6**
nde_Latn	46.7	**52.9**	ndo_Latn	47.3	**51.2**	nds_Latn	39.1	**48.6**	nep_Deva	70.7	**72.5**
ngu_Latn	41.5	**44.1**	nld_Latn	**62.7**	62.1	nmf_Latn	43.1	**47.8**	nnb_Latn	38.5	**46.0**
nno_Latn	63.1	**65.5**	nob_Latn	**60.6**	60.0	nor_Latn	**61.7**	60.8	npi_Deva	69.7	**70.1**
nse_Latn	43.9	**45.1**	nso_Latn	**53.6**	52.4	nya_Latn	55.4	**61.9**	nyn_Latn	43.6	**47.0**
nyy_Latn	31.1	**37.7**	nzi_Latn	34.4	**38.3**	ori_Orya	**70.3**	69.5	ory_Orya	**71.4**	69.3
oss_Cyrl	48.0	**57.6**	ote_Latn	35.6	**35.7**	pag_Latn	52.0	**54.3**	pam_Latn	40.1	**45.4**
pan_Guru	**67.8**	66.4	pap_Latn	65.5	**66.2**	pau_Latn	42.0	**43.3**	pcm_Latn	63.8	**67.1**
pdt_Latn	58.1	**58.7**	pes_Arab	**70.5**	69.6	pis_Latn	67.5	**67.9**	pls_Latn	46.5	**49.1**
plt_Latn	**52.6**	50.1	poh_Latn	47.2	**48.2**	pol_Latn	64.8	**68.8**	pon_Latn	53.1	**53.9**
por_Latn	68.0	**72.3**	prk_Latn	**56.4**	**56.4**	prs_Arab	**68.9**	**68.9**	pxm_Latn	**40.6**	40.2
qub_Latn	58.3	**59.1**	quc_Latn	51.0	**53.3**	qug_Latn	64.5	**68.2**	quh_Latn	62.4	**68.6**
quw_Latn	53.5	**55.6**	quy_Latn	**71.6**	70.6	quz_Latn	64.8	**67.6**	qvi_Latn	63.2	**65.2**
rap_Latn	47.9	**49.4**	rar_Latn	45.8	**52.7**	rmy_Latn	45.0	**47.9**	ron_Latn	59.4	**66.7**
rop_Latn	57.0	**58.1**	rug_Latn	51.2	**55.1**	run_Latn	48.2	**53.6**	rus_Cyrl	69.4	**72.3**
sag_Latn	44.7	**47.1**	sah_Cyrl	59.0	**61.6**	sba_Latn	38.7	**41.0**	seh_Latn	47.5	**49.9**
sin_Sinh	**67.5**	66.6	slk_Latn	60.3	**60.6**	slv_Latn	62.4	**62.6**	sme_Latn	36.8	**48.8**
smo_Latn	56.2	**61.6**	sna_Latn	40.9	**45.4**	snd_Arab	67.7	**68.7**	som_Latn	33.6	**35.8**
sop_Latn	34.1	**39.0**	sot_Latn	45.2	**48.6**	spa_Latn	64.2	**67.5**	sqi_Latn	**70.8**	70.3
srm_Latn	48.7	**52.4**	srn_Latn	64.0	**64.4**	srp_Latn	64.9	**69.0**	ssw_Latn	38.0	**46.9**
sun_Latn	54.0	**56.6**	suz_Deva	58.2	**60.3**	swe_Latn	67.8	**69.0**	swh_Latn	62.1	**64.2**
sxn_Latn	49.0	**51.8**	tam_Taml	72.6	**73.7**	tat_Cyrl	65.5	**68.0**	tbz_Latn	36.0	**42.7**
tca_Latn	42.2	**48.3**	tdt_Latn	58.2	**66.0**	tel_Telu	70.9	**71.4**	teo_Latn	24.0	**26.6**
tgk_Cyrl	64.0	**65.7**	tgl_Latn	57.5	**58.8**	tha_Thai	65.3	**65.5**	tih_Latn	56.6	**60.3**
tir_Ethi	51.9	**52.5**	tlh_Latn	64.0	**65.3**	tob_Latn	**44.3**	44.0	toh_Latn	38.1	**40.3**
toi_Latn	39.5	**49.4**	toj_Latn	36.7	**39.2**	ton_Latn	48.2	**50.9**	top_Latn	22.9	**26.6**
tpi_Latn	68.4	**69.7**	tpm_Latn	45.8	**51.4**	tsn_Latn	45.6	**46.5**	tsz_Latn	37.3	**42.8**
tuc_Latn	56.3	**62.2**	tui_Latn	45.6	**48.1**	tuk_Latn	56.9	**63.5**	tum_Latn	48.1	**50.1**
tur_Latn	62.6	**65.8**	twi_Latn	42.1	**47.5**	tyv_Cyrl	58.2	**63.0**	tzh_Latn	38.8	**44.2**
tzo_Latn	38.3	**42.5**	udm_Cyrl	53.8	**54.2**	ukr_Cyrl	65.3	**67.5**	urd_Arab	**61.3**	60.4
uzb_Latn	**59.7**	58.0	uzn_Cyrl	65.3	**65.8**	ven_Latn	43.4	**46.3**	vie_Latn	**70.0**	69.3
wal_Latn	42.1	**49.0**	war_Latn	45.6	**52.4**	wbm_Latn	**57.5**	56.1	wol_Latn	34.0	**40.4**
xav_Latn	29.9	**33.1**	xho_Latn	45.3	**48.5**	yan_Latn	51.2	**53.6**	yao_Latn	40.0	**46.5**
yap_Latn	39.3	**42.0**	yom_Latn	36.2	**37.8**	yor_Latn	**47.4**	46.9	yua_Latn	36.7	**39.8**
yue_Hani	58.4	**60.4**	zai_Latn	39.3	**44.2**	zho_Hani	64.6	**67.0**	zlm_Latn	**70.3**	69.7
zom_Latn	47.2	**49.9**	zsm_Latn	**69.7**	68.0	zul_Latn	50.6	**53.4**			

Table 11: F1 scores of models on **Taxi1500** (Part II).

Language	Baseline	LANGSAMP	Language	Baseline	LANGSAMP	Language	Baseline	LANGSAMP	Language	Baseline	LANGSAMP
ace_Latn	71.5	**73.6**	acm_Arab	82.2	**83.0**	afr_Latn	82.3	**82.7**	ajp_Arab	**83.4**	81.8
aka_Latn	62.2	**67.2**	als_Latn	82.4	**84.4**	amh_Ethi	**74.2**	73.6	apc_Arab	**83.9**	82.9
arb_Arab	**83.8**	82.9	ary_Arab	**81.5**	80.2	arz_Arab	**84.5**	84.1	asm_Beng	83.6	**84.2**
ast_Latn	**88.4**	88.0	ayr_Latn	51.1	**53.8**	azb_Arab	71.5	**74.7**	azj_Latn	87.0	**88.0**
bak_Cyrl	84.6	**86.6**	bam_Latn	**47.9**	47.6	ban_Latn	80.3	**83.0**	bel_Cyrl	**83.7**	83.4
bem_Latn	63.0	**63.9**	ben_Beng	83.3	**84.3**	bjn_Latn	77.1	**78.5**	bod_Tibt	**73.5**	69.2
bos_Latn	86.5	**88.2**	bul_Cyrl	86.1	**87.5**	cat_Latn	84.8	**86.4**	ceb_Latn	81.8	**84.6**
ces_Latn	**89.1**	86.9	cjk_Latn	46.6	**48.1**	ckb_Arab	**83.9**	80.2	crh_Latn	74.0	**76.2**
cym_Latn	**75.9**	75.4	dan_Latn	86.8	**87.4**	deu_Latn	86.5	**87.8**	dyu_Latn	42.6	**44.5**
dzo_Tibt	68.7	**72.6**	ell_Grek	79.5	**80.0**	eng_Latn	**90.8**	90.0	epo_Latn	**83.8**	82.2
est_Latn	80.6	**81.6**	eus_Latn	82.1	**82.2**	ewe_Latn	49.3	**51.5**	fao_Latn	83.7	**84.9**
fij_Latn	56.1	**58.0**	fin_Latn	82.1	**82.9**	fon_Latn	41.7	**44.6**	fra_Latn	87.9	**89.6**
fur_Latn	77.6	**80.2**	gla_Latn	**57.6**	54.3	gle_Latn	62.2	**64.1**	glg_Latn	87.8	**89.0**
grn_Latn	**75.0**	74.5	guj_Gujr	83.9	**84.7**	hat_Latn	77.4	**79.1**	hau_Latn	**62.7**	62.1
heb_Hebr	77.9	**79.2**	hin_Deva	84.1	**84.4**	hne_Deva	77.9	**80.1**	hrv_Latn	87.3	**89.0**
hun_Latn	86.8	**87.6**	hye_Armn	**83.0**	82.5	ibo_Latn	72.3	**74.1**	ilo_Latn	75.8	**79.6**
ind_Latn	88.7	**89.1**	isl_Latn	78.5	**79.1**	ita_Latn	87.7	**89.2**	jav_Latn	80.2	**80.3**
jpn_Jpan	87.1	**87.9**	kab_Latn	31.1	**36.9**	kac_Latn	49.3	**52.3**	kam_Latn	49.1	**49.5**
kan_Knda	**83.2**	82.0	kat_Geor	81.8	**83.7**	kaz_Cyrl	84.2	**84.9**	kbp_Latn	**45.1**	44.2
kea_Latn	75.4	**77.0**	khm_Khmr	84.3	**84.4**	kik_Latn	57.1	**59.9**	kin_Latn	69.5	**70.5**
kir_Cyrl	**80.7**	80.3	kmb_Latn	48.2	**49.5**	kmr_Latn	**70.7**	70.0	kon_Latn	65.3	**69.2**
kor_Hang	**85.2**	83.9	lao_Laoo	**85.1**	84.2	lij_Latn	77.7	**79.6**	lim_Latn	74.7	**75.2**
lin_Latn	69.3	**71.4**	lit_Latn	**86.5**	84.7	lmo_Latn	77.7	**79.1**	ltz_Latn	76.6	**79.1**
lua_Latn	**59.1**	56.4	lug_Latn	55.5	**59.1**	luo_Latn	52.6	**53.0**	lus_Latn	65.3	**67.9**
lvs_Latn	**84.4**	83.6	mai_Deva	83.4	**84.0**	mal_Mlym	**80.6**	79.9	mar_Deva	**84.1**	82.5
min_Latn	77.7	**79.6**	mkd_Cyrl	83.3	**84.6**	mlt_Latn	82.9	**83.0**	mos_Latn	44.9	**46.6**
mri_Latn	54.4	**59.3**	mya_Mymr	80.1	**81.6**	nld_Latn	**86.5**	85.8	nno_Latn	**86.6**	86.4
nob_Latn	85.8	**86.1**	npi_Deva	**86.8**	86.0	nso_Latn	61.3	**61.9**	nya_Latn	71.1	**72.7**
oci_Latn	83.1	**84.9**	ory_Orya	79.7	**80.3**	pag_Latn	78.7	**79.7**	pan_Guru	77.4	**79.0**
pap_Latn	77.2	**79.0**	pes_Arab	87.6	**89.2**	plt_Latn	68.4	**68.5**	pol_Latn	86.4	**86.7**
por_Latn	87.3	**88.6**	prs_Arab	85.8	**88.4**	quy_Latn	63.7	**64.0**	ron_Latn	**86.4**	84.5
run_Latn	**68.3**	67.2	rus_Cyrl	87.6	**87.9**	sag_Latn	52.4	**55.1**	san_Deva	**77.9**	77.8
sat_Olck	53.0	**57.4**	scn_Latn	77.6	**78.2**	sin_Sinh	**84.5**	84.1	slk_Latn	86.1	**87.0**
slv_Latn	**86.4**	85.5	smo_Latn	73.4	**74.1**	sna_Latn	**59.3**	58.0	snd_Arab	72.1	**76.9**
som_Latn	**61.8**	59.8	sot_Latn	65.3	**67.6**	spa_Latn	**86.4**	86.2	srd_Latn	74.0	**75.8**
srp_Cyrl	**85.8**	85.2	ssw_Latn	67.5	**68.1**	sun_Latn	84.0	**85.2**	swe_Latn	86.6	**87.3**
swh_Latn	76.0	**78.6**	szl_Latn	74.3	**75.5**	tam_Taml	80.6	**84.3**	tat_Cyrl	84.0	**85.2**
tel_Telu	85.3	**85.7**	tgk_Cyrl	**81.6**	80.9	tgl_Latn	81.9	**83.0**	tha_Thai	87.4	**88.9**
tir_Ethi	59.9	**61.4**	tpi_Latn	80.6	**82.3**	tsn_Latn	**59.1**	55.2	tso_Latn	59.3	**61.2**
tuk_Latn	**78.3**	78.2	tum_Latn	70.3	**70.8**	tur_Latn	82.9	**83.6**	twi_Latn	61.4	**68.0**
uig_Arab	77.7	**80.0**	ukr_Cyrl	**84.7**	84.5	umb_Latn	**45.9**	45.8	urd_Arab	81.3	**81.9**
vec_Latn	**82.0**	81.1	vie_Latn	84.9	**85.8**	war_Latn	81.7	**83.4**	wol_Latn	49.2	**52.1**
xho_Latn	62.4	**64.0**	yor_Latn	46.6	**51.8**	zsm_Latn	**87.2**	86.6	zul_Latn	**73.8**	73.6

Table 12: F1 scores of models on **SIB200**.

Language	Baseline	LangSAMP	Language	Baseline	LangSAMP	Language	Baseline	LangSAMP	Language	Baseline	LangSAMP
ace_Latn	**42.8**	42.6	afr_Latn	76.4	**76.8**	als_Latn	81.7	**82.2**	amh_Ethi	**40.8**	39.0
ara_Arab	55.9	**56.5**	arg_Latn	78.9	**80.0**	arz_Arab	56.8	**58.4**	asm_Beng	**66.0**	64.8
ast_Latn	82.3	**84.5**	aym_Latn	**45.9**	43.8	aze_Latn	65.4	**65.9**	bak_Cyrl	**61.3**	60.2
bar_Latn	68.2	**70.0**	bel_Cyrl	74.4	**74.6**	ben_Beng	**71.1**	70.5	bih_Deva	**55.9**	55.2
bod_Tibt	27.1	**35.0**	bos_Latn	**73.0**	72.6	bre_Latn	**64.6**	63.9	bul_Cyrl	**75.6**	75.2
cat_Latn	**83.8**	**83.8**	cbk_Latn	**53.9**	51.6	ceb_Latn	54.0	**57.5**	ces_Latn	78.6	**78.7**
che_Cyrl	45.6	**55.7**	chv_Cyrl	**77.4**	75.2	ckb_Arab	73.2	**73.3**	cos_Latn	59.1	**59.7**
crh_Latn	**51.9**	50.2	csb_Latn	61.9	**62.0**	cym_Latn	**62.2**	60.2	dan_Latn	**81.7**	**81.7**
deu_Latn	76.0	**76.7**	diq_Latn	51.7	**53.4**	div_Thaa	48.2	**55.2**	ell_Grek	**73.0**	72.9
eml_Latn	42.0	**42.1**	eng_Latn	**83.5**	83.4	epo_Latn	68.1	**69.0**	est_Latn	72.9	**73.6**
eus_Latn	57.5	**59.1**	ext_Latn	45.0	**46.2**	fao_Latn	**70.9**	69.6	fas_Arab	**52.7**	51.5
fin_Latn	75.0	**75.4**	fra_Latn	76.4	**76.8**	frr_Latn	**55.3**	54.5	fry_Latn	**77.2**	76.3
fur_Latn	**57.5**	57.1	gla_Latn	59.5	**65.4**	gle_Latn	72.7	**72.8**	glg_Latn	79.9	**80.4**
grn_Latn	54.0	**54.1**	guj_Gujr	58.8	**59.2**	hbs_Latn	62.7	**65.5**	heb_Hebr	50.4	**51.7**
hin_Deva	68.4	**68.9**	hrv_Latn	77.1	**77.8**	hsb_Latn	74.2	**75.9**	hun_Latn	76.2	**76.7**
hye_Armn	54.1	**57.2**	ibo_Latn	**57.3**	**57.3**	ido_Latn	79.0	**79.9**	ilo_Latn	73.6	**76.8**
ina_Latn	57.8	**58.5**	ind_Latn	62.2	**64.0**	isl_Latn	**72.4**	72.1	ita_Latn	78.4	**78.6**
jav_Latn	**58.9**	57.8	jbo_Latn	25.0	**25.1**	jpn_Jpan	**21.0**	17.5	kan_Knda	58.5	**58.7**
kat_Geor	67.5	**68.2**	kaz_Cyrl	50.0	**51.1**	khm_Khmr	42.4	**43.1**	kin_Latn	63.5	**69.0**
kir_Cyrl	**45.4**	44.2	kor_Hang	**52.7**	51.2	ksh_Latn	58.0	**59.2**	kur_Latn	63.1	**64.8**
lat_Latn	**75.7**	74.7	lav_Latn	72.4	**74.9**	lij_Latn	42.7	**46.3**	lim_Latn	70.3	**71.0**
lin_Latn	**51.5**	49.2	lit_Latn	**75.0**	74.6	lmo_Latn	**75.3**	74.1	ltz_Latn	68.6	**69.0**
lzh_Hani	**14.6**	11.5	mal_Mlym	**62.8**	61.7	mar_Deva	**63.0**	62.0	mhr_Cyrl	60.6	**61.9**
min_Latn	41.5	**42.9**	mkd_Cyrl	75.7	**76.3**	mlg_Latn	57.8	**58.7**	mlt_Latn	66.9	**71.3**
mon_Cyrl	66.0	**68.9**	mri_Latn	**49.5**	48.0	msa_Latn	68.7	**69.0**	mwl_Latn	48.9	**51.1**
mya_Mymr	**53.8**	53.6	mzn_Arab	44.9	**47.4**	nan_Latn	83.6	**85.0**	nap_Latn	59.0	**59.3**
nds_Latn	**77.8**	77.0	nep_Deva	**63.1**	61.7	nld_Latn	**81.4**	81.1	nno_Latn	76.5	**77.3**
nor_Latn	76.6	**77.7**	oci_Latn	68.6	**70.7**	ori_Orya	31.7	**31.9**	oss_Cyrl	**53.5**	51.3
pan_Guru	**49.7**	48.1	pms_Latn	78.9	**80.3**	pnb_Arab	**65.1**	64.7	pol_Latn	78.0	**78.1**
por_Latn	77.6	**78.7**	pus_Arab	40.5	**42.0**	que_Latn	66.1	**67.4**	roh_Latn	**62.0**	58.8
ron_Latn	**76.0**	75.6	rus_Cyrl	69.0	**69.1**	sah_Cyrl	**75.3**	69.3	san_Deva	35.5	**37.1**
scn_Latn	65.4	**66.0**	sco_Latn	84.5	**87.3**	sgs_Latn	60.7	**64.0**	sin_Sinh	**53.0**	49.4
slk_Latn	**77.9**	77.3	slv_Latn	**80.2**	80.1	snd_Arab	**44.3**	41.9	som_Latn	52.2	**55.8**
spa_Latn	74.0	**76.5**	sqi_Latn	**77.7**	77.6	srp_Cyrl	63.5	**64.3**	sun_Latn	**55.4**	53.8
swa_Latn	**69.1**	69.0	swe_Latn	70.3	**73.0**	szl_Latn	68.1	**70.8**	tam_Taml	54.9	**55.3**
tat_Cyrl	**65.9**	63.0	tel_Telu	**50.2**	49.1	tgk_Cyrl	63.0	**66.0**	tgl_Latn	75.5	**76.8**
tha_Thai	**4.6**	2.2	tuk_Latn	56.3	**56.5**	tur_Latn	76.1	**76.6**	uig_Arab	47.8	**48.6**
ukr_Cyrl	**76.6**	**76.6**	urd_Arab	**66.2**	65.1	uzb_Latn	73.1	**74.5**	vec_Latn	**68.6**	67.7
vep_Latn	71.1	**71.2**	vie_Latn	73.0	**73.1**	vls_Latn	75.5	**76.5**	vol_Latn	59.6	**59.7**
war_Latn	66.4	**66.6**	wuu_Hani	**32.3**	28.5	xmf_Geor	63.6	**65.7**	yid_Hebr	50.4	**55.7**
yor_Latn	60.7	**61.6**	yue_Hani	**23.7**	21.6	zea_Latn	**66.8**	63.7	zho_Hani	**24.7**	20.9

Table 13: F1 scores of models on **NER**.

Language	Baseline	LangSAMP	Language	Baseline	LangSAMP	Language	Baseline	LangSAMP	Language	Baseline	LangSAMP
afr_Latn	87.8	**88.2**	ajp_Arab	**70.4**	69.2	aln_Latn	**52.9**	50.7	amh_Ethi	65.9	**67.0**
ara_Arab	**66.5**	**66.5**	bam_Latn	**42.2**	41.5	bel_Cyrl	**86.0**	85.5	ben_Beng	**83.2**	83.0
bre_Latn	**61.0**	60.0	bul_Cyrl	88.1	**88.2**	cat_Latn	86.5	**86.7**	ceb_Latn	**65.2**	**65.2**
ces_Latn	**84.7**	**84.7**	cym_Latn	**65.7**	64.6	dan_Latn	**90.4**	**90.4**	deu_Latn	**87.9**	87.6
ell_Grek	**86.1**	84.9	eng_Latn	**96.0**	**96.0**	est_Latn	**83.7**	**83.7**	eus_Latn	**65.2**	64.2
fao_Latn	**88.8**	88.3	fas_Arab	71.3	**71.9**	fin_Latn	**82.1**	81.7	fra_Latn	**86.2**	85.8
gla_Latn	**57.9**	57.5	gle_Latn	**64.0**	**64.0**	glg_Latn	83.1	**83.5**	glv_Latn	**51.7**	51.0
grc_Grek	**71.7**	71.0	grn_Latn	19.7	**21.3**	gsw_Latn	79.7	**80.0**	hbo_Hebr	34.0	**38.6**
heb_Hebr	**68.5**	68.0	hin_Deva	70.4	**71.8**	hrv_Latn	**85.6**	85.5	hsb_Latn	**83.2**	82.9
hun_Latn	81.5	**82.3**	hye_Armn	84.0	**84.4**	hyw_Armn	81.2	**81.4**	ind_Latn	**83.6**	83.3
isl_Latn	**82.9**	82.7	ita_Latn	87.8	**88.7**	jav_Latn	73.4	**74.0**	jpn_Jpan	22.6	**28.9**
kaz_Cyrl	**76.5**	76.0	kmr_Latn	74.1	**74.2**	kor_Hang	**52.8**	52.2	lat_Latn	**72.8**	71.7
lav_Latn	**83.6**	**83.6**	lij_Latn	**76.4**	75.6	lit_Latn	**81.5**	81.3	lzh_Hani	**23.7**	21.3
mal_Mlym	**86.4**	86.1	mar_Deva	**81.4**	80.7	mlt_Latn	79.5	**80.0**	myv_Cyrl	64.4	**64.6**
nap_Latn	70.6	**75.9**	nds_Latn	77.0	**77.8**	nld_Latn	88.2	**88.3**	nor_Latn	**88.2**	**88.2**
pcm_Latn	**57.3**	**57.3**	pol_Latn	**83.5**	83.2	por_Latn	88.0	**88.1**	quc_Latn	**61.1**	58.8
ron_Latn	81.4	**81.7**	rus_Cyrl	**88.9**	88.5	sah_Cyrl	**74.0**	73.0	san_Deva	**25.7**	24.2
sin_Sinh	56.0	**56.5**	slk_Latn	84.3	**84.9**	slv_Latn	**77.2**	77.0	sme_Latn	73.0	**73.1**
spa_Latn	87.6	**87.8**	sqi_Latn	76.3	**76.9**	srp_Latn	**85.3**	**85.3**	swe_Latn	92.5	**92.6**
tam_Taml	73.6	**73.7**	tat_Cyrl	70.2	**71.1**	tel_Telu	**81.8**	81.7	tgl_Latn	**75.5**	75.0
tha_Thai	55.9	**56.7**	tur_Latn	**71.3**	71.1	uig_Arab	**68.2**	**68.2**	ukr_Cyrl	85.0	**85.2**
urd_Arab	62.0	**64.6**	vie_Latn	**68.3**	67.4	wol_Latn	**61.6**	59.5	xav_Latn	**11.8**	10.5
yor_Latn	62.3	**62.6**	yue_Hani	37.4	**41.3**	zho_Hani	38.7	**44.8**			

Table 14: F1 scores of models on **POS**.

Language	English	Closest donor	Language	English	Closest donor	Language	English	Closest donor	Language	English	Closest donor
ace_Latn	**63.3**	60.1	ach_Latn	35.6	**48.1**	acr_Latn	**48.8**	46.7	afr_Latn	**58.6**	58.5
ahk_Latn	5.4	**8.3**	aka_Latn	**44.9**	41.2	aln_Latn	**56.2**	54.7	als_Latn	**57.1**	57.1
alz_Latn	34.1	**43.0**	aoj_Latn	40.9	**46.2**	arb_Arab	**55.4**	55.4	arn_Latn	43.1	**44.4**
arz_Arab	33.7	**40.3**	asm_Beng	53.4	**61.5**	ayr_Latn	52.7	**62.2**	azb_Arab	**61.0**	61.0
bak_Cyrl	54.7	**59.7**	bam_Latn	48.9	**55.6**	ban_Latn	**43.0**	42.5	bar_Latn	**47.8**	43.3
bci_Latn	34.6	**37.1**	bcl_Latn	54.2	**60.5**	bel_Cyrl	59.1	**61.5**	bem_Latn	44.2	**49.4**
bhw_Latn	**50.2**	46.9	bim_Latn	47.3	**55.1**	bis_Latn	**68.4**	68.1	bqc_Latn	33.2	**41.6**
btx_Latn	**56.7**	53.8	bul_Cyrl	62.5	**62.6**	bum_Latn	39.6	**42.2**	bzj_Latn	**65.7**	60.3
cac_Latn	43.8	**46.0**	cak_Latn	51.0	**57.9**	caq_Latn	42.7	**51.0**	cat_Latn	61.2	**62.3**
cce_Latn	**43.8**	38.0	ceb_Latn	**49.8**	49.1	ces_Latn	63.3	**63.7**	cfm_Latn	**58.3**	57.1
chk_Latn	**42.8**	38.9	chv_Cyrl	60.3	**64.3**	ckb_Arab	58.3	**67.0**	cmn_Hani	60.8	**73.0**
crh_Cyrl	61.4	**67.7**	crs_Latn	62.3	**63.5**	csy_Latn	**58.3**	56.7	ctd_Latn	**56.6**	55.8
cuk_Latn	39.1	**40.8**	cym_Latn	**51.9**	46.0	dan_Latn	**58.1**	54.0	deu_Latn	**51.5**	51.5
dln_Latn	**54.4**	54.4	dtp_Latn	51.5	**51.6**	dyu_Latn	**55.6**	48.2	dzo_Tibt	50.6	**58.1**
ell_Grek	**56.9**	53.9	eng_Latn	**78.0**	78.0	enm_Latn	**70.8**	67.0	epo_Latn	**58.3**	58.3
eus_Latn	**25.2**	21.4	ewe_Latn	46.4	**52.1**	fao_Latn	56.5	**64.8**	fas_Arab	69.6	**70.2**
fil_Latn	56.7	**58.7**	fin_Latn	**56.4**	55.7	fon_Latn	**36.8**	35.4	fra_Latn	**66.8**	66.8
gaa_Latn	36.9	**47.7**	gil_Latn	40.4	**47.2**	giz_Latn	48.4	**48.5**	gkn_Latn	**40.0**	34.1
gla_Latn	**45.6**	45.6	gle_Latn	41.8	**45.1**	glv_Latn	37.3	**48.7**	gom_Latn	34.8	**41.6**
guc_Latn	**39.6**	37.6	gug_Latn	39.0	**46.0**	guj_Gujr	67.1	**70.4**	gur_Latn	37.0	**44.2**
gya_Latn	39.6	**41.8**	gym_Latn	45.4	**52.9**	hat_Latn	**63.0**	60.0	hau_Latn	54.0	**59.6**
heb_Hebr	**16.7**	15.2	hif_Latn	42.4	**53.6**	hil_Latn	**63.7**	61.6	hin_Deva	**64.8**	64.8
hne_Deva	64.1	**67.5**	hnj_Latn	61.5	**63.2**	hra_Latn	48.2	**53.1**	hrv_Latn	**62.7**	60.7
hun_Latn	65.2	**65.9**	hus_Latn	37.6	**40.7**	hye_Armn	67.2	**69.3**	iba_Latn	57.9	**59.2**
ifa_Latn	49.7	**51.5**	ifb_Latn	**48.3**	48.1	ikk_Latn	46.6	**52.5**	ilo_Latn	**58.8**	55.7
isl_Latn	53.5	**61.2**	ita_Latn	62.8	**67.1**	ium_Latn	51.4	**58.0**	ixl_Latn	36.6	**38.2**
jam_Latn	**66.1**	61.0	jav_Latn	43.9	**47.6**	jpn_Jpan	**58.6**	58.6	kaa_Latn	57.7	**62.6**
kac_Latn	44.5	**47.3**	kal_Latn	31.5	**34.5**	kan_Knda	60.6	**67.5**	kat_Geor	55.2	**62.2**
kbp_Latn	34.9	**39.5**	kek_Latn	**41.5**	40.3	khm_Khmr	**64.7**	64.7	kia_Latn	48.0	**51.7**
kin_Latn	47.2	**52.5**	kir_Cyrl	61.1	**64.7**	kjb_Latn	44.7	**48.1**	kjh_Cyrl	**52.3**	51.1
kmr_Cyrl	45.5	**53.1**	knv_Latn	**42.6**	40.5	kor_Hang	69.8	**71.3**	kpg_Latn	**64.1**	57.4
kri_Latn	**63.2**	56.0	ksd_Latn	54.2	**54.4**	kss_Latn	16.2	**21.6**	ksw_Mymr	**50.4**	50.3
lam_Latn	34.7	**35.6**	lao_Laoo	69.1	**72.7**	lat_Latn	57.2	**62.9**	lav_Latn	**60.4**	57.7
leh_Latn	**43.5**	37.2	lhu_Latn	22.3	**29.0**	lin_Latn	47.1	**54.7**	lit_Latn	58.3	**59.7**
ltz_Latn	**48.2**	48.2	lug_Latn	**46.1**	39.0	luo_Latn	40.6	**41.2**	lus_Latn	**51.6**	51.6
mad_Latn	55.3	**63.0**	mah_Latn	**41.6**	38.3	mai_Deva	**62.7**	60.5	mam_Latn	**33.9**	33.2
mau_Latn	5.5	**8.4**	mbb_Latn	52.6	**53.1**	mck_Latn	**41.9**	41.2	mcn_Latn	37.7	**39.3**
mdy_Ethi	51.6	**57.6**	meu_Latn	54.9	**55.8**	mfe_Latn	66.0	**66.2**	mgh_Latn	30.3	**33.1**
mhr_Cyrl	36.0	**38.5**	min_Latn	**49.9**	40.7	miq_Latn	**52.2**	52.2	mkd_Cyrl	**71.2**	70.3
mlt_Latn	**50.7**	50.7	mos_Latn	40.3	**41.2**	mps_Latn	**57.1**	53.1	mri_Latn	50.9	**52.6**

Table 15: F1 scores of LANGSAMP on **Taxi1500** using English and the closest donor language as source (Part I).

Language	English	Closest donor	Language	English	Closest donor	Language	English	Closest donor	Language	English	Closest donor
msa_Latn	41.7	**42.0**	mwm_Latn	**55.1**	55.0	mxv_Latn	**29.6**	27.4	mya_Mymr	**54.4**	53.4
mzh_Latn	39.7	**45.1**	nan_Latn	31.5	**31.8**	naq_Latn	41.7	**43.7**	nav_Latn	21.1	**29.5**
nch_Latn	**44.0**	36.6	ncj_Latn	38.6	**39.1**	ndc_Latn	34.7	**36.6**	nde_Latn	45.7	**49.8**
nds_Latn	**49.6**	44.0	nep_Deva	68.0	**72.1**	ngu_Latn	43.4	**48.2**	nld_Latn	**61.1**	53.7
nnb_Latn	40.7	**46.1**	nno_Latn	**63.1**	63.1	nob_Latn	57.2	**58.2**	nor_Latn	56.4	**57.8**
nse_Latn	45.9	**48.5**	nso_Latn	**48.6**	48.6	nya_Latn	**56.0**	47.4	nyn_Latn	43.0	**44.1**
nzi_Latn	33.0	**33.8**	ori_Orya	**67.3**	67.3	ory_Orya	66.9	**70.7**	oss_Cyrl	55.5	**57.5**
pag_Latn	**55.5**	52.5	pam_Latn	**42.0**	37.8	pan_Guru	**64.1**	64.1	pap_Latn	**65.6**	59.8
pcm_Latn	**66.1**	65.9	pdt_Latn	**60.0**	56.5	pes_Arab	**69.0**	69.0	pis_Latn	64.3	**65.0**
plt_Latn	46.8	**52.9**	poh_Latn	44.3	**45.5**	pol_Latn	64.8	**65.1**	pon_Latn	50.5	**52.2**
prk_Latn	52.9	**53.0**	prs_Arab	69.2	**70.0**	pxm_Latn	34.5	**41.5**	qub_Latn	51.5	**56.3**
qug_Latn	**65.0**	61.3	quh_Latn	**66.7**	58.8	quw_Latn	55.9	**56.0**	quy_Latn	65.5	**67.7**
qvi_Latn	**62.0**	58.5	rap_Latn	48.9	**49.3**	rar_Latn	48.9	**51.9**	rmy_Latn	45.4	**49.1**
rop_Latn	**56.6**	54.7	rug_Latn	53.8	**55.1**	run_Latn	48.0	**55.2**	rus_Cyrl	**68.1**	68.1
sah_Cyrl	55.1	**57.6**	sba_Latn	39.1	**41.4**	seh_Latn	45.0	**46.7**	sin_Sinh	64.1	**66.9**
slv_Latn	**63.8**	60.7	sme_Latn	**42.8**	37.6	smo_Latn	**60.8**	54.2	sna_Latn	42.6	**44.9**
som_Latn	33.9	**35.5**	sop_Latn	**36.4**	36.0	sot_Latn	43.5	**45.5**	spa_Latn	**64.2**	64.2
srm_Latn	48.1	**48.4**	srn_Latn	**63.7**	62.8	srp_Latn	64.9	**65.2**	ssw_Latn	**43.7**	37.7
suz_Deva	**58.0**	57.8	swe_Latn	**66.8**	65.3	swh_Latn	**59.8**	59.8	sxn_Latn	**46.6**	40.2
tat_Cyrl	62.2	**68.2**	tbz_Latn	36.4	**39.5**	tca_Latn	43.3	**50.3**	tdt_Latn	**60.3**	55.1
teo_Latn	**23.7**	23.1	tgk_Cyrl	**60.9**	60.9	tgl_Latn	56.7	**58.7**	tha_Thai	**63.8**	63.8
tir_Ethi	**50.1**	50.1	tlh_Latn	**65.0**	65.0	tob_Latn	43.3	**50.4**	toh_Latn	37.1	**39.0**
toj_Latn	**36.6**	34.1	ton_Latn	47.3	**51.5**	top_Latn	**21.9**	21.3	tpi_Latn	63.8	**67.6**
tsn_Latn	39.8	**44.1**	tsz_Latn	40.4	**41.0**	tuc_Latn	**57.4**	56.9	tui_Latn	**43.7**	43.7
tum_Latn	**47.6**	43.2	tur_Latn	**62.1**	62.1	twi_Latn	**41.4**	38.9	tyv_Cyrl	59.8	**60.3**
tzo_Latn	**39.5**	39.5	udm_Cyrl	49.6	**49.9**	ukr_Cyrl	**62.4**	62.2	uzb_Latn	53.5	**57.7**
ven_Latn	41.9	**48.6**	vie_Latn	62.4	**65.4**	wal_Latn	**48.9**	42.7	war_Latn	47.7	**54.5**
wol_Latn	**37.2**	33.9	xav_Latn	**25.5**	23.7	xho_Latn	**44.9**	44.4	yan_Latn	50.3	**53.5**
yap_Latn	42.8	**42.9**	yom_Latn	**37.6**	34.1	yor_Latn	**41.8**	35.4	yua_Latn	40.1	**43.2**
zai_Latn	**42.6**	41.4	zho_Hani	**60.7**	60.7	zlm_Latn	**68.4**	65.5	zom_Latn	**44.6**	44.4
zul_Latn	51.9	**52.2**									

Table 16: F1 scores of LANGSAMP on **Taxi1500** using English and the closest donor language as source (Part II).

Language	English	Closest donor	Language	English	Closest donor	Language	English	Closest donor	Language	English	Closest donor
ace_Latn	69.9	**72.4**	acm_Arab	80.6	**81.4**	afr_Latn	81.4	**81.8**	ajp_Arab	81.4	**83.0**
als_Latn	**82.3**	82.3	amh_Ethi	**72.6**	72.6	apc_Arab	81.7	**83.2**	arb_Arab	**81.5**	81.5
arz_Arab	82.1	**84.4**	asm_Beng	**83.0**	83.0	ast_Latn	87.1	**87.6**	ayr_Latn	48.6	**51.1**
azj_Latn	**86.5**	84.0	bak_Cyrl	84.3	**86.5**	bam_Latn	**46.5**	42.2	ban_Latn	79.5	**81.3**
bem_Latn	**61.1**	51.4	ben_Beng	83.7	**84.0**	bjn_Latn	75.9	**77.9**	bod_Tibt	65.7	**71.0**
bul_Cyrl	86.3	**86.6**	cat_Latn	**85.7**	85.2	ceb_Latn	81.2	**83.2**	ces_Latn	**86.3**	85.6
ckb_Arab	**80.0**	76.8	crh_Latn	**76.8**	75.7	cym_Latn	73.6	**76.6**	dan_Latn	85.0	**86.0**
dyu_Latn	**43.6**	42.4	dzo_Tibt	**68.2**	59.8	ell_Grek	**79.5**	78.8	eng_Latn	**88.9**	88.9
est_Latn	**78.9**	78.1	eus_Latn	78.8	**80.7**	ewe_Latn	**49.9**	46.7	fao_Latn	**84.4**	83.6
fin_Latn	80.9	**81.5**	fon_Latn	**40.8**	38.1	fra_Latn	**87.8**	87.8	fur_Latn	77.4	**77.9**
gle_Latn	61.5	**64.4**	glg_Latn	**87.6**	87.6	grn_Latn	71.6	**73.2**	guj_Gujr	82.1	**83.4**
hau_Latn	59.3	**64.2**	heb_Hebr	76.8	**80.2**	hin_Deva	**82.8**	82.8	hne_Deva	77.9	**79.5**
hun_Latn	86.6	**87.5**	hye_Armn	**81.3**	80.3	ibo_Latn	**71.4**	71.3	ilo_Latn	76.1	**76.7**
isl_Latn	78.0	**78.3**	ita_Latn	86.4	**87.5**	jav_Latn	**79.9**	79.7	jpn_Jpan	**86.8**	86.8
kac_Latn	**48.9**	46.6	kam_Latn	45.8	**48.3**	kan_Knda	82.9	**83.0**	kat_Geor	**83.7**	81.0
kbp_Latn	**42.8**	42.2	kea_Latn	**73.1**	73.1	khm_Khmr	**82.7**	82.7	kik_Latn	55.1	**56.7**
kir_Cyrl	79.3	**80.1**	kmb_Latn	**46.2**	42.6	kmr_Latn	**69.8**	68.9	kon_Latn	**65.2**	63.4
lao_Laoo	**83.4**	82.9	lij_Latn	**76.4**	74.9	lim_Latn	**74.1**	73.0	lin_Latn	68.2	**73.3**
lmo_Latn	77.0	**78.3**	ltz_Latn	**76.4**	76.4	lua_Latn	**54.4**	54.3	lug_Latn	**58.2**	55.8
lus_Latn	**64.8**	64.8	lvs_Latn	**83.2**	83.0	mai_Deva	**82.9**	82.1	mal_Mlym	**79.8**	79.3
min_Latn	76.7	**79.8**	mkd_Cyrl	**83.6**	82.8	mlt_Latn	**81.3**	81.3	mos_Latn	**44.7**	40.9
mya_Mymr	**80.5**	78.8	nld_Latn	85.1	**86.4**	nno_Latn	**86.0**	86.0	nob_Latn	**84.8**	84.4
nso_Latn	**57.6**	57.6	nya_Latn	69.2	**70.9**	oci_Latn	**85.0**	84.1	ory_Orya	78.6	**79.0**
pan_Guru	**76.4**	76.4	pap_Latn	76.9	**78.1**	pes_Arab	**87.5**	87.3	plt_Latn	67.5	**69.3**
por_Latn	85.3	**86.8**	prs_Arab	85.0	**85.5**	quy_Latn	**62.6**	59.7	ron_Latn	84.0	**84.4**
rus_Cyrl	**86.8**	86.8	sag_Latn	**51.3**	50.2	san_Deva	72.9	**76.6**	sat_Olck	**56.4**	53.5
sin_Sinh	**82.7**	82.7	slk_Latn	**85.4**	85.1	slv_Latn	84.2	**87.4**	smo_Latn	74.2	**75.3**
snd_Arab	**70.4**	70.4	som_Latn	58.9	**61.1**	sot_Latn	**64.1**	63.2	spa_Latn	**84.4**	84.4
srp_Cyrl	84.8	**85.0**	ssw_Latn	64.1	**65.2**	sun_Latn	82.6	**85.2**	swe_Latn	84.2	**86.2**
szl_Latn	**72.4**	72.4	tam_Taml	**81.2**	81.2	tat_Cyrl	**83.6**	83.6	tel_Telu	84.0	**85.4**
tgl_Latn	**82.1**	81.7	tha_Thai	85.4	**85.7**	tir_Ethi	**60.3**	60.3	tpi_Latn	**80.3**	75.7
tso_Latn	57.3	**60.3**	tuk_Latn	78.1	**78.5**	tum_Latn	65.4	**68.5**	tur_Latn	**80.4**	80.4
uig_Arab	**75.5**	75.5	ukr_Cyrl	**84.3**	83.8	umb_Latn	41.0	**46.5**	urd_Arab	79.1	**80.6**
vie_Latn	**86.2**	83.9	war_Latn	80.7	**81.3**	wol_Latn	**50.5**	46.4	xho_Latn	**60.1**	59.8
zho_Hans	**89.6**	89.2	zho_Hant	**88.8**	88.8	zsm_Latn	**86.4**	86.0	zul_Latn	68.1	**69.8**

Table 17: F1 scores of LANGSAMP on **SIB200**. using English and the closest donor language as source.

Language	English	Closest donor	Language	English	Closest donor	Language	English	Closest donor	Language	English	Closest donor
ace_Latn	41.5	**56.9**	afr_Latn	75.8	**80.3**	als_Latn	**80.9**	80.9	amh_Ethi	**39.7**	39.7
arg_Latn	82.2	**88.8**	arz_Arab	55.1	**82.6**	asm_Beng	**69.0**	45.9	ast_Latn	84.6	**85.8**
aze_Latn	65.0	**74.0**	bak_Cyrl	62.5	**72.2**	bar_Latn	**68.2**	62.8	bel_Cyrl	74.9	**79.7**
bih_Deva	56.2	**67.6**	bod_Tibt	35.2	**35.7**	bos_Latn	70.1	**75.2**	bre_Latn	63.3	**66.0**
cat_Latn	83.8	**85.1**	cbk_Latn	**53.7**	48.9	ceb_Latn	**56.0**	26.8	ces_Latn	**77.9**	69.6
chv_Cyrl	73.6	**84.3**	ckb_Arab	**76.0**	60.6	cos_Latn	**63.0**	61.9	crh_Latn	52.7	**59.4**
cym_Latn	61.7	**62.1**	dan_Latn	**81.4**	81.3	deu_Latn	**74.6**	74.6	diq_Latn	54.0	**72.2**
ell_Grek	71.9	**72.0**	eml_Latn	**41.3**	41.3	eng_Latn	**83.5**	83.5	epo_Latn	**68.3**	68.3
eus_Latn	60.9	**65.1**	ext_Latn	44.2	**48.6**	fao_Latn	68.7	**79.2**	fas_Arab	**55.0**	53.6
fra_Latn	**76.5**	76.5	frr_Latn	**52.0**	52.0	fry_Latn	**74.6**	73.9	fur_Latn	**58.2**	54.0
gle_Latn	**72.6**	69.6	glg_Latn	80.7	**86.1**	grn_Latn	55.1	**59.8**	guj_Gujr	**61.2**	61.0
heb_Hebr	52.0	**52.9**	hin_Deva	**69.4**	69.4	hrv_Latn	77.2	**79.8**	hsb_Latn	**74.3**	69.7
hye_Armn	53.0	**62.2**	ibo_Latn	58.1	**58.4**	ido_Latn	**82.6**	81.5	ilo_Latn	**80.0**	74.9
ind_Latn	**67.6**	67.6	isl_Latn	70.1	**75.4**	ita_Latn	78.2	**79.5**	jav_Latn	56.0	**86.4**
jpn_Jpan	**22.0**	22.0	kan_Knda	57.5	**61.8**	kat_Geor	**68.7**	60.1	kaz_Cyrl	50.5	**57.1**
kin_Latn	**69.6**	67.3	kir_Cyrl	44.3	**60.9**	kor_Hang	50.4	**51.2**	ksh_Latn	**59.7**	51.4
lat_Latn	71.9	**81.4**	lav_Latn	**74.4**	69.0	lij_Latn	45.2	**54.2**	lim_Latn	**69.3**	61.2
lit_Latn	74.2	**76.1**	lmo_Latn	**73.6**	65.5	ltz_Latn	**67.9**	67.9	lzh_Hani	**14.8**	14.8
mar_Deva	62.5	**76.6**	mhr_Cyrl	60.6	**72.3**	min_Latn	42.6	**57.5**	mkd_Cyrl	72.2	**73.1**
mlt_Latn	**75.9**	75.9	mon_Cyrl	**68.7**	60.9	mri_Latn	**50.0**	47.0	msa_Latn	67.6	**73.0**
mya_Mymr	55.3	**56.3**	mzn_Arab	43.3	**47.2**	nan_Latn	**88.1**	36.6	nap_Latn	**63.0**	55.3
nep_Deva	56.9	**60.4**	nld_Latn	**80.8**	80.0	nno_Latn	**77.6**	77.6	nor_Latn	77.9	**80.4**
ori_Orya	**34.2**	34.2	oss_Cyrl	50.6	**59.1**	pan_Guru	**51.5**	51.5	pms_Latn	**80.9**	78.4
pol_Latn	**77.7**	71.1	por_Latn	78.9	**84.9**	pus_Arab	42.6	**45.3**	que_Latn	**70.4**	55.5
ron_Latn	**77.8**	75.5	rus_Cyrl	**67.5**	67.5	sah_Cyrl	71.9	**77.9**	san_Deva	38.4	**53.4**
sco_Latn	**86.4**	84.5	sgs_Latn	66.4	**69.8**	sin_Sinh	**53.0**	51.2	slk_Latn	**76.4**	55.9
snd_Arab	**41.8**	41.8	som_Latn	**57.5**	56.2	spa_Latn	**77.6**	77.6	sqi_Latn	76.8	**78.7**
sun_Latn	50.8	**75.1**	swa_Latn	**71.8**	71.8	swe_Latn	**70.9**	65.8	szl_Latn	**70.9**	70.9
tat_Cyrl	63.8	**76.5**	tel_Telu	48.1	**49.0**	tgk_Cyrl	**68.4**	68.4	tgl_Latn	71.9	**73.7**
tuk_Latn	54.4	**57.3**	tur_Latn	**77.1**	77.1	uig_Arab	47.7	**62.3**	ukr_Cyrl	76.6	**85.3**
uzb_Latn	73.2	**76.0**	vec_Latn	68.0	**75.1**	vep_Latn	**72.0**	63.0	vie_Latn	**72.3**	49.7
vol_Latn	**61.0**	36.5	war_Latn	**64.9**	56.1	wuu_Hani	35.7	**66.7**	xmf_Geor	**69.3**	55.7
yor_Latn	**69.3**	41.7	yue_Hani	25.7	**73.5**	zea_Latn	62.9	**75.4**	zho_Hani	**25.2**	25.2

Table 18: F1 scores of LANGSAMP on **NER** using English and the closest donor language as source.

Language	English	Closest donor	Language	English	Closest donor	Language	English	Closest donor	Language	English	Closest donor
afr_Latn	**88.5**	79.5	ajp_Arab	**71.1**	41.9	aln_Latn	**53.4**	45.1	amh_Ethi	**66.8**	66.8
bam_Latn	**43.0**	31.2	bel_Cyrl	86.4	**93.8**	ben_Beng	**87.5**	80.2	bre_Latn	61.1	**62.3**
cat_Latn	86.8	**95.8**	ceb_Latn	**66.7**	32.5	ces_Latn	**85.4**	73.3	cym_Latn	**65.5**	60.4
deu_Latn	**88.2**	88.2	ell_Grek	**84.9**	75.5	eng_Latn	**96.0**	96.0	est_Latn	**84.7**	77.4
fao_Latn	**88.7**	67.5	fas_Arab	**72.2**	69.1	fin_Latn	**82.2**	75.8	fra_Latn	**85.8**	85.8
gle_Latn	64.6	**65.5**	glg_Latn	83.6	**87.8**	glv_Latn	51.9	**57.8**	grc_Grek	**71.6**	71.6
gsw_Latn	**82.7**	82.7	hbo_Hebr	**38.9**	37.4	heb_Hebr	67.9	**69.3**	hin_Deva	**77.2**	77.2
hsb_Latn	**83.7**	73.4	hun_Latn	**82.2**	42.0	hye_Armn	**85.1**	84.9	hyw_Armn	**83.0**	56.8
isl_Latn	**82.7**	81.2	ita_Latn	88.9	**92.4**	jav_Latn	75.4	**78.8**	jpn_Jpan	**33.1**	33.1
kmr_Latn	**76.6**	61.6	kor_Hang	**52.7**	45.3	lat_Latn	72.8	**74.2**	lav_Latn	**83.7**	78.4
lit_Latn	**82.1**	80.7	lzh_Hani	**24.5**	24.5	mal_Mlym	**86.0**	52.1	mar_Deva	**84.1**	81.7
myv_Cyrl	**65.9**	58.4	nap_Latn	**82.4**	70.6	nds_Latn	**79.1**	34.0	nld_Latn	**88.2**	82.2
pcm_Latn	**58.2**	48.1	pol_Latn	84.2	**89.1**	por_Latn	87.9	**92.0**	quc_Latn	**63.3**	52.6
rus_Cyrl	**88.7**	88.7	sah_Cyrl	74.2	**74.5**	san_Deva	25.5	**32.7**	sin_Sinh	**56.2**	34.4
slv_Latn	77.6	**79.0**	sme_Latn	**74.8**	60.6	spa_Latn	**87.8**	87.8	sqi_Latn	**77.5**	72.7
swe_Latn	**92.7**	83.2	tam_Taml	**74.6**	74.6	tat_Cyrl	**72.4**	70.9	tel_Telu	**80.9**	55.9
tha_Thai	**58.3**	27.5	tur_Latn	**71.2**	71.2	uig_Arab	**68.2**	48.3	ukr_Cyrl	85.6	**91.7**
vie_Latn	**68.4**	32.4	wol_Latn	**61.6**	57.4	xav_Latn	**16.7**	11.2	yor_Latn	**62.7**	46.5
zho_Hani	**47.4**	47.4									

Table 19: F1 scores of LANGSAMP on **POS** using English and the closest donor language as source.